RESEARCH PAPER • FIELD DYNAMICS • MODIFIED GRAVITY

Distributed Field Processing: The Gravitational Constitutive Law

(No Dark Matter Required)

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ABSTRACT

We derive the MOND interpolating function $\mu(x) = x/(1+x)$ from first principles. For forty years, this function has organized galactic phenomenology around a single acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s², yet its theoretical origin remained unknown. We prove it is the *unique* solution within the Minimal Single-Channel Saturating Closure class — a conditional theorem, not a phenomenological fit.

The derivation proceeds from the *field closure function* (FC_Σ): the function by which a field object couples to its gravitational environment (the NET Field) at the closure boundary Σ. Two physical requirements constrain this coupling — it must be lossless (no wake, no drag, no hysteresis) and locally evaluable (causality forbids global computation). These constraints, formalized as Postulates A1–A3, uniquely determine the constitutive law.

The framework extends to relativistic regimes. A fourth postulate (operational calibration of rods and clocks) derives the 1PN metric structure, yielding $\Phi = \Psi$ and gravitational slip $\gamma = 1$. The optical-mechanical analog recovers full gravitational lensing ($\theta = 4GM/bc^2$) from two-sphere field closure, resolving the historical "half-lensing" problem of scalar theories. The scalar Lagrangian $L(x) = \frac{1}{2}x^2 - x + \ln(1+x)$ is derived, establishing the variational foundation.

The framework reframes dark matter phenomenology: "missing mass" is phantom density — the field's nonlinear response to baryons, misread through Newtonian analysis. Falsifiable predictions distinguish this framework from particle dark matter: fast galactic bars ($\mathcal{R} \approx 1.0$–1.4), external field effects in satellite galaxies, baryon-only lensing predictions in clusters, and wide-binary anomalies at $g < a_0$.

Distributed Field Processing | Field Dynamics | Field Closure | Field Object | NET Field | AQUAL | MOND | Dark Matter Alternative | Modified Gravity | Full-State Snapshot (FSS)

Scope & Commitments

Theoretical results (derived in this paper):

- **Constitutive law:** $\mu(x) = x/(1+x)$ — uniquely determined from closure constraints (Postulates A1–A3); a theorem, not a fit
- **Scalar Lagrangian:** $L(x) = \frac{1}{2}x^2 - x + \ln(1+x)$ — the variational foundation
- **1PN metric:** $\Phi = \Psi$, $\gamma = 1$ — derived from operational calibration (Postulate A4)
- **Full gravitational lensing:** $\theta = 4GM/bc^2$ — via optical-mechanical analog
- **Phantom density:** "missing mass" as operator residual, not substance

Falsifiable predictions:

- Fast galactic bars ($\mathscr{R} \approx 1.0$–1.4) — no halo-wake friction
- External field effects in satellites — environmental dependence CDM lacks
- Baryon-only cluster lensing — Bullet Cluster forward-modeling pipeline
- Wide-binary anomalies at $g < a_0$

Relativistic status: This paper establishes *covariant completion at 1PN* (metric structure, lensing, $\gamma = 1$). What remains is the *fully covariant action* — the $S = \int \mathscr{L}\sqrt{-g}\, d^4x$ that generates time-dependent dynamics (gravitational waves, propagation). The static/1PN framework here constrains any such action.

Cosmology: CMB and structure formation require relativistic completion beyond this paper's scope. However, existing work (Skordis & Złośnik 2021) demonstrates that MOND-type theories *can* fit the CMB; DFP constrains what the galactic limit of any such completion must look like.

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Appendix A: Formal Lemmas

1. Introduction

1.1 The Observational Puzzles

The mass discrepancy in galactic dynamics is one of the central problems in astrophysics. Across five decades of galaxy mass and morphological type, observations reveal a consistent pattern: gravitational effects exceed what visible matter can produce. This discrepancy manifests in multiple, independent ways:

Rotation curves remain flat. In galaxy after galaxy, stars at the outer edge orbit just as fast as stars much closer to the center. Newtonian mechanics predicts velocities should decline with distance from the center, much like planets in our solar system, where distant planets orbit more slowly than inner ones. Galactic rotation curves do not decline. They remain flat, or even rise, far beyond the visible disk.

Gravitational lensing exceeds visible mass. Light bending around galaxies and clusters implies gravitational pulls 5–10 times stronger than the visible matter can produce. The inferred "extra" mass is not optional; it is required by the geometry of spacetime itself.

Scaling relations are remarkably tight. The Baryonic Tully–Fisher Relation (BTFR) connects a galaxy's total baryonic mass to its asymptotic rotation velocity: $M_b \propto V^4$. This relation holds across five decades in mass with remarkably little scatter. If the "missing mass" were an independent substance distributed arbitrarily, why would it correlate so precisely with visible matter?

Cluster dynamics require something extra. Galaxy clusters move as if they contain far more mass than observed in stars and hot gas. The temperature of intracluster gas implies deep gravitational wells, deeper than visible matter can explain.

These observations demand explanation. The dominant hypothesis, particle dark matter, posits a new substance that interacts gravitationally but not electromagnetically. Yet an alternative phenomenology exists: Milgrom's Modified Newtonian Dynamics (MOND), which modifies the gravitational response below a characteristic acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s². MOND successfully predicts the tight BTFR, the shape of rotation curves, and the radial acceleration relation across a wide range of galaxy types, without invoking new particles.

1.2 The Gap: μ(x) is Not Derived

The AQUAL formulation (Aquadratic Lagrangian) captures MOND physics through a modified Poisson equation:

$$\nabla \cdot \left[\mu \left(\frac{|\nabla \Phi|}{a_0} \right) \nabla \Phi \right] = 4\pi G \rho$$

Here Φ is the gravitational potential, ρ is the baryonic density, and $\mu(x)$ is the *interpolating function* that governs the transition between Newtonian and modified regimes. The function must satisfy:

- $\mu(x) \to 1$ as $x \to \infty$: recovers Newton in strong fields
- $\mu(x) \to x$ as $x \to 0$: produces MOND phenomenology in weak fields

But these asymptotic constraints do not uniquely determine $\mu(x)$. The "simple" form $\mu(x) = x/(1+x)$, the "standard" form $\mu(x) = x/\sqrt{1+x^2}$, and infinitely many other functions all satisfy both limits. This is not a minor ambiguity; it is the central theoretical gap in modified gravity.

For forty years, $\mu(x)$ has been chosen by curve-fitting, not derived from principle. The theory lacks a foundation that would uniquely specify which $\mu(x)$ is correct, and why. **This paper closes that gap.** Section 2 identifies the missing physics; Section 3 derives the unique $\mu(x)$.

1.3 This Paper's Contribution

This paper closes the gap identified above. We derive $\mu(x) = x/(1+x)$ as a theorem, not a fit. The derivation proceeds from physical constraints that any gravitational coupling must satisfy, requires no free functional choices, and yields a unique result.

The logical structure. Section 2 establishes the architecture: matter couples to the gravitational field, and this coupling must produce a definite result. Two empirical facts constrain how this can work. First, we observe no wakes, drag, or hysteresis as objects move through gravitational fields, so coupling must be lossless. Second, causality forbids signals traveling faster than light, so coupling must be locally evaluable. Section 3 formalizes these as Postulates A1–A3 and proves that $\mu(x) = x/(1+x)$ is the unique function satisfying them. Section 4 verifies the result against galactic observations.

Beyond galactic dynamics. The framework extends to relativistic regimes. Section 8.2 derives the 1PN metric structure from a fourth postulate (operational calibration of rods and clocks), yielding $\Phi = \Psi$ and gravitational slip $\gamma = 1$. The optical-mechanical analog recovers full gravitational lensing ($\theta = 4GM/bc^2$), resolving the historical "half-lensing" problem of scalar theories without invoking additional fields.

Dark matter reframed. The framework reinterprets "missing mass" as phantom density: the discrepancy between nonlinear field dynamics and linear Newtonian inference. This is not an independent substance but the field's response to baryons. The phantom density tracks baryonic structure because it is determined by it.

Testable predictions. Four predictions distinguish this framework from particle dark matter: fast galactic bars (Section 7.1), external field effects in satellite galaxies (Section 4.3), baryon-only lensing predictions in clusters (Section 8.5), and wide-binary anomalies (Section 7.3). These are quantitative, falsifiable, and accessible to current observations.

Terminology. We name the components of gravitational coupling: the matter configuration (field object), the combined gravitational environment (NET Field), and the interface where coupling resolves (closure boundary). These are descriptive labels for structure that effective field theories typically leave implicit. A mapping to standard EFT language appears in §8.7.

> **Reading Paths**
>
> **For the derivation:** Readers who want the mathematics may proceed directly to Section 3. The derivation stands on its own; it requires only the postulates stated there.
>
> **For the mechanism:** Sections 2, 5, and 6 develop the conceptual framework. These sections explain *why* the derivation works by examining what happens when fields couple. A running example (a nickel-iron asteroid) makes the concepts concrete throughout.
>
> Both paths arrive at the same result: $\mu(x) = x/(1+x)$.

2. Field Closure Architecture

When an asteroid moves through the Solar System, it exists within a gravitational environment: the Sun's field, the planets' contributions, the galaxy's background. The asteroid's mass and this environment must become compatible. Something governs how they couple into one consistent state.

This "something" is not new physics; it is what fields already do. We call it **field closure** and give it a name (FC_Σ) so we can reason about it precisely. This section builds the foundation: how objects interact with gravitational fields, what closure requires, and why these requirements uniquely determine $\mu(x)$.

A note on terminology: "Field closure" is not a new mechanism we are proposing. It is a name for what must happen when any two field configurations couple: they must become compatible, settle into a consistent state, and register a result. Every field theory has this. We are naming it so we can ask: what does compatibility *require*? The answer is $\mu(x)$.

Scope note: The constitutive law we derive embeds into the AQUAL field equation, an effective field theory for equilibrated systems: the field equation depends on the current mass distribution, not its history. This is the standard approach for galaxy dynamics. Section 5.5 discusses the deeper nature of field closure; Section 8.7 addresses the scope and limitations of this non-relativistic framework.

2.1 The NET Field

A galaxy doesn't experience one gravitational field. It experiences dozens: its cluster's field, the local group's field, the cosmic web's field. Yet it responds as if there's only one. Why?

Consider the layers a galaxy sits in:

- Its cluster's gravitational field
- The local group's field
- The cosmic background

These contributions don't arrive separately. At the galaxy's location, they combine into one gravitational environment: the **NET Field**.

What matters is local. The galaxy doesn't need to reconstruct the universe's history. It doesn't need to know about the Virgo cluster's detailed structure, the local group's formation, or the cosmic web's evolution. Everything "upstream" has already contributed to the local field state. What the galaxy encounters is the field as it is *here, now*.

Think of a river. Upstream there are bends, rocks, tributaries, but the boulder in the stream experiences only the current at its location. It doesn't need to know the river's history. The NET Field works the same way: all upstream contributions are already encoded in the local flow. **What matters is local.**

→ **Next question:** Now that we know what the object *encounters* (the NET Field), we need to specify what the object *is* as a gravitational source.

2.2 The Field Object

An asteroid contains roughly 10^{33} atoms. Each atom is a quantum field. Together, they don't form a list of separate particles; they form **one unified configuration** bound together by electromagnetic forces. This is a **field object**: many quantum fields coupled into a single correlated state.

Think of a musical chord. A chord isn't three separate notes that happen to sound together; the chord IS the relationship between the notes. Change one note and you have a different chord. A field object works the same way. The iron atoms, the crystalline lattice, the binding relationships between them: these define one thing. The correlations are as real as the atoms themselves.

This matters because you cannot couple to part of a field object. The NET Field doesn't interact with each atom separately; it couples to the asteroid as a unified whole.

Mass is field count. A gold nugget has more mass than a foam ball of the same size. Why? The gold contains more coupled fields in the same volume. Denser materials pack more quantum fields into the same space. When we say "more massive," we mean "more fields in the configuration." An asteroid (~10^{33} fields) has more than a pebble (~10^{24} fields); a star (~10^{57} fields) has more than an asteroid.

Dual role: Every field object is both embedded in the NET Field (experiences external gravity) and a source (contributes to the NET Field elsewhere). Our asteroid is embedded in the Sun's field AND creates a field that other objects experience.

One more property is essential before we examine how field objects and the NET Field interface:

> **Coupling Is Continuous**
>
> As a field object moves through the NET Field, it doesn't couple once and coast. The field object and NET Field must continuously settle into a consistent state. This is not a one-time handshake; it is ongoing.
>
> If coupling is continuous, so is field closure. FC_Σ operates continuously as the field object moves. The gravitational field doesn't lag behind; it follows because closure is continuously complete.

→ **Next question:** Now we know what objects encounter (NET Field) and what they are (field objects). Where do they meet, and how do they interface?

2.3 Closure: Where They Meet and How They Interface

To describe the interaction between field object and NET Field, we identify:

Question	Answer	Term
Where do they meet?	Geometric locus of constraint enforcement	**Closure boundary (Σ)**
What is the field closure function?	Constraint functional governing coupling	**Field closure (FC_Σ)**
What is the process?	Reaching the fixed point	**Settlement**
What is the result?	Recorded boundary state	**Registration (B^*_Σ)**

The closure boundary (Σ) is the geometric locus where the field object meets the NET Field. It is where the coupling constraint is enforced, the stage on which field closure operates. Σ is not owned by either field; it is the neutral interface where they must achieve consistency.

The field closure function (FC_Σ) governs coupling at Σ. When the NET Field couples to a field object, FC_Σ enforces compatibility and drives **settlement** to a unique **registered state**. Field closure is not a third physical field; it is the function that resolves underdetermination in the coupled field description.

Multiple field contributions must resolve to ONE consistent local state. At any point: one clock rate. One ruler. One causal speed *c*. **Counterfactual:** Could a single point have two different clock rates simultaneously? No. Physics would contradict itself. The contributions must *close* into a single calibration state.

The **registered state** (B^*_Σ) is the output of settlement: the boundary data at Σ that determines the **settled potential** Φ* throughout the domain. Gradients of this potential are gravity, lensing, and time dilation: three manifestations of the same underlying field dynamics.

Two analogies clarify the structure of field closure:

The chord constraint. When three piano notes sound together, they form a chord. But the chord isn't just the notes; it's the *relationship* between them, the harmonic structure that makes a C-major sound different from a D-minor. You cannot "hear half a chord": either all three notes are present with their relationships, or the chord doesn't exist. Similarly, a field object cannot be partially closed. The closure constraint applies to the entire configuration simultaneously: the relationships between components define the entity, not the components themselves.

Transactional atomicity. In any consistent accounting system, whether a financial ledger or a database, an entry completes fully or not at all. Partial completion introduces inconsistency. Field closure inherits this structure: the scope is the entire field object, and it resolves completely. There is no "half-closed" field state.

The Canonical Chain

Field Object + NET Field

(inputs)

↓

Closure Boundary (Σ)

(where)

↓

FC_Σ: Field Closure Function

READ → SETTLE

↓

B^*_Σ: Registration

(settlement complete)

This is THE mechanism. At the closure boundary Σ, FC_Σ operates: both configurations share the same local geometry, enabling a lossless read, and the two fields settle into a consistent state. The output B^*_Σ is the registered boundary state. Gradients $\nabla\ell(x)$ are what we experience as gravity.

One mechanism, three manifestations: Lensing is not separate from gravity. Time dilation is not separate from lensing. They are all gradients of the same calibration field, the output of closure.

→ **Next question:** Closure must work, but HOW must it work? What quality requirements does observation impose?

2.4 The Constraints: Lossless and Local

Observation imposes two requirements on how closure must work.

Constraint 1: Lossless (Full-State Snapshot). The entire field object must be registered, not a coarse-grained summary. **Counterfactual:** What if closure were lossy? Observable consequences would include:

- **Wakes**: constraint debris left behind by moving objects
- **Drag**: the field resisting motion
- **Hysteresis**: history-dependent gravitational response

- **Secular drift**: accumulated errors over cosmic time

We observe NONE of these. Gravitational field dynamics are conservative, history-independent, and wake-free. **Conclusion:** Full-State Snapshot (FSS) is empirically required. It is not a theoretical preference but an observational constraint on any viable theory of field dynamics. This is not an assumption; it's forced by observation.

Constraint 2: Local. Closure must be locally evaluable, with no global computation and no instantaneous synchronization. The field object and NET Field couple because they share the same local conditions at Σ. Compatibility is automatic: both configurations exist in the same local field state, so no global coordination is needed.

Counterfactual: What if closure required global computation? If gravity depended on knowing the state of a distant galaxy in real time, signals would need to travel faster than light. This is inadmissible.

Locality ensures compatibility: At Σ, the field object and NET Field exist together in the same local conditions. Large-scale structure contributes to the NET Field, but this contribution is already present locally. Closure happens locally because both configurations are co-located; they settle into a consistent state through their shared local conditions, without requiring global coordination. This is why DFP is "distributed."

→ **Next question:** We now know closure must be local, lossless, and produce ONE calibration state. What equation satisfies these requirements?

2.5 The Complete Picture

With all the concepts established, we can now trace the complete mechanism through a single example.

The Asteroid: Tracing the Full Mechanism

Picture a nickel-iron asteroid in our solar system. It is not merely 10^{33} atoms; it is 10^{33} *fields*: iron atoms with their binding fields, crystallized silica with its lattice modes, all coupled together into a single unified configuration. This is a **field object**: a quantum many-body configuration, a unified field chord where the relationships between components define the entity.

What does this asteroid encounter? The Sun's gravitational field, the planets' contributions, the Milky Way's background field. Like water in a river, much has happened upstream, but what the asteroid meets is the stream as it is *now*, *here*. This combined environment is the **NET Field**.

Where do they meet? At the **closure boundary (Σ)**, where the asteroid's field object and the NET Field exist together in the same local conditions. They share the same local field state, so they are compatible by virtue of co-location. **Field closure** governs this coupling, enforcing compatibility at Σ and driving **settlement** to a unique **registered state**.

The NET Field cannot couple to "part" of the asteroid's field object. You cannot play half a chord; the relationships between notes define the sound. Similarly, the NET Field must couple to the *whole* unified configuration. This is **snapshot accounting**: lossless, high-fidelity coupling of the entire correlated chord. Because both configurations exist in the same local conditions at Σ, the coupling can be complete. As the asteroid tumbles through the solar system, it leaves no "constraint debris" behind. Its gravitational influence follows it cleanly. If closure were lossy, the asteroid's path would be littered with gravitational "ghosts" marking where it used to be. We would see them. We don't.

Continuous coupling in action: As the asteroid tumbles through the solar system, it continuously settles with the NET Field, not once. Its gravitational influence doesn't lag behind; it follows because settlement is continuously complete. This is why the field is static and history-independent: the current state is always fully resolved.

Coherent coupling of a 10^{33}-field chord under continuous interaction scales with its coupling load. This appears in the effective theory as the **local geometry scale** $\ell(x)$ that specifies local clock rates and length scales. Gradients $\nabla\ell(x)$ are what we experience as gravity.

$$\text{Field Object} + \text{NET Field} \rightarrow \Sigma \rightarrow \text{FC}_\Sigma(\text{READ} \rightarrow \text{SETTLE}) \rightarrow \text{B}^*_\Sigma$$

The asteroid is not "pulled" by a force at a distance. It is embedded in a field; that field couples to its field object; $\ell(x)$ describes the existing local geometry and FC_Σ is where coupling happens. Gradients $\nabla\ell(x)$ determine motion. **Gravity is the macroscopic signature of uncompromising field accounting.**

2.6 From Architecture to Field Law

We now know: closure must be local, lossless, and produce ONE calibration state. What equation satisfies these requirements?

Requirement	Source	Mathematical Implication
Local evaluation	§2.4	Differential equation, not integral
Single closure channel	§2.2, §2.3	Single scalar μ (field object is unified → one coupling channel)
Lossless (FSS)	§2.4	Detailed balance (reversible, no information loss)
Newton recovery	Empirical	$\mu \to 1$ as $x \to \infty$
Modified weak-field	Empirical	$\mu \neq 1$ as $x \to 0$
Conservation	Basic physics	Divergence form

The minimal form satisfying these requirements is:

$$\nabla \cdot [\mu(x) \nabla \Phi] = 4\pi G \rho$$

where $x = |\nabla \Phi|/a_0$ and $\mu(x)$ must satisfy the architecture constraints listed above.

> **Section 2 Summary: The Five Architectural Facts**
>
> 1. **Fact 1: One NET Field.** Objects sit in one combined gravitational environment
> 2. **Fact 2: One field object.** Objects are unified configurations; internal correlations define the entity
> 3. **Fact 3: One closure boundary (Σ).** Where field object meets NET Field; the stage for constraint enforcement
> 4. **Fact 4: Lossless.** Closure must be FSS-complete (no wakes, drag, hysteresis)
> 5. **Fact 5: Local.** Field dynamics must be locally evaluable (no faster-than-c coordination)
>
> These five facts are not assumptions. They are derived from observation and physical consistency. Together, they constrain the architecture of gravitational field dynamics and determine what $\mu(x)$ can be.
>
> **These five facts constrain μ(x) to a unique form. Section 3 derives it.**

Section 3 derives the unique $\mu(x)$ that satisfies these requirements.

3. The Derivation

With the field closure architecture established (§2), we now derive the interpolating function $\mu(x)$. The goal is to derive its precise form from physical constraints, not to fit it from data. The logical status of each claim is tracked in the Claim Ledger (§3.6).

3.1 From Newton to the Gap

Newtonian gravity emerges from a single, elegant equation, the Poisson equation:

$$\nabla^2 \Phi_N = 4\pi G \rho$$

where Φ_N is the gravitational potential and ρ is the mass density. This linear partial differential equation permits superposition: the potential from multiple sources is the sum of individual potentials. Combined with the equation of motion $\ddot{\mathbf{r}} = -\nabla \Phi_N$, it yields the inverse-square force law that governs planetary orbits and falling apples alike.

At galactic scales, however, this framework fails. Observed rotation curves remain flat far beyond the visible disk, implying gravitational accelerations that exceed Newtonian predictions by factors of 5–10. The standard resolution posits invisible "dark matter" halos; an alternative is to modify the gravitational field equation itself.

The AQUAL Generalization

Bekenstein & Milgrom (1984) introduced AQUAL (AQUAdratic Lagrangian), which replaces the linear Poisson equation with a nonlinear generalization:

$$\nabla \cdot \left[\mu\left(\frac{|\nabla \Phi|}{a_0} \right) \nabla \Phi \right] = 4\pi G \rho$$

Here $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ is the MOND acceleration scale, and $\mu(x)$ is an *interpolating function* that governs the transition between two regimes:

- **High accelerations** ($|\nabla\Phi| \gg a_0$): $\mu(x) \to 1$, recovering Newtonian gravity.
- **Low accelerations** ($|\nabla\Phi| \ll a_0$): $\mu(x) \to x$, producing the deep-MOND dynamics that explain flat rotation curves.

The key insight is that AQUAL derives from a Lagrangian (action principle), which guarantees conservation of energy, momentum, and angular momentum, properties that naive "force law" modifications of MOND can violate.

Mathematical character: This is an *elliptic* partial differential equation, a boundary-value problem, not a wave equation. Solutions are determined by the snapshot configuration $\rho(\mathbf{x})$ and boundary conditions; there are no time derivatives and no velocity dependence. This does not imply instantaneous physical propagation; it is the quasi-static limit of an underlying causal theory.

Local transmissivity. Let $\mathbf{g} \equiv -\nabla\Phi$. Define the **effective field**:

$$\mathbf{g}_{\mathrm{res}} \equiv \mu\left(\frac{|\mathbf{g}|}{a_0} \right) \mathbf{g}$$

The AQUAL equation is then a **Gauss law for $\mathbf{g}_{\mathrm{res}}$**: $\nabla \cdot \mathbf{g}_{\mathrm{res}} = -4\pi G\rho$. Integrating over any closed surface:

$$\oint_S \mathbf{g}_{\text{res}} \cdot d\mathbf{A} = -4\pi G M_{\text{enclosed}}$$

The constitutive factor μ acts as a local transmissivity (a constitutive weight) on the raw gradient field. By Gauss's theorem, the surface integral expresses a **weighted-flux invariance**: for any closed surface enclosing the same mass, the flux of $\mathbf{g}_{\text{res}}$ is invariant.

The Gap: μ(x) is Undetermined

AQUAL specifies the *structure* of the field equation but leaves the interpolating function $\mu(x)$ as a free choice. Any function satisfying the two asymptotic limits will produce flat rotation curves; the precise form is traditionally selected by fitting observational data. The "simple" form $\mu = x/(1+x)$, the "standard" form $\mu = x/\sqrt{1+x^2}$, and infinitely many other functions all satisfy the boundary conditions. AQUAL works, but we do not know *why* one $\mu(x)$ rather than another.

What This Paper Provides

We close this gap. The central result of this work is that $\mu(x)$ is *not* a free function. It is uniquely determined within the minimal rational class by three physical and methodological postulates:

1. **Newtonian Correspondence**: Recovery of standard gravity at high accelerations.
2. **Distributed Field Processing**: The field possesses a finite capacity to encode gravitational gradients, saturating at scale a_0.
3. **Single-Channel Saturation**: The field operates as a single-channel processor; the equilibrium partition between resolved and unresolved states follows from detailed balance.

These three constraints exactly saturate the three parameters of the minimal rational class, yielding $\mu(x) = x/(1+x)$ as the *unique* solution within that class. The interpolating function is not fitted; it is **derived from the postulates**.

The remainder of this section formalizes these postulates and proves the uniqueness theorem.

3.2 The Three Postulates

Following Einstein's insight that "there is no space empty of field," we treat the gravitational medium simply as *the field* throughout this work, the fundamental entity possessing resolution properties, not an empty container.

The theory rests upon three postulates concerning the field's capacity to encode gravitational information. These are not arbitrary assumptions but physical constraints that any resolution-limited medium must satisfy.

We define the gravitational acceleration vector $\mathbf{g} \equiv -\nabla\Phi$ and the dimensionless gradient intensity $x \equiv |\mathbf{g}|/a_0$, where $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ is the fundamental resolution scale.

POSTULATE 1: NEWTONIAN CORRESPONDENCE

At high accelerations, gravity is Newtonian. When the gravitational gradient greatly exceeds the resolution scale ($x \gg 1$), the field is fully resolved, and the constitutive response approaches unity:

$$\lim_{x\to\infty} \mu(x) = 1$$

Physical Motivation for Resolution Limits

The hypothesis that physical systems possess finite resolution capacity is not novel; it appears throughout modern physics. Heisenberg's uncertainty principle establishes that phase space cannot be resolved below $\hbar$. The Bekenstein bound limits information content in any region to $S \leq 2\pi RE/\hbar c$, implying that spacetime itself has finite informational capacity. Holographic arguments suggest that gravitational degrees of freedom scale with area, not volume.

In this context, a gravitational resolution floor is not merely plausible but expected. The scale a_0 is of order cH_0 (up to $\mathcal{O}(1)$ factors), coincident with the cosmic expansion rate, hinting at cosmological origin.[†] The field cannot encode gravitational structure finer than the horizon scale permits.

[†] *Cosmic coincidence note:* The numerical proximity $a_0 \sim cH_0$ suggests a cosmological origin for the resolution scale. This could mean either (i) a_0 tracks a horizon scale $H(z)$ that evolves with cosmic time, or (ii) a_0 is tied more directly to a late-time de Sitter scale (e.g., Λ) rather than the instantaneous Hubble rate. Distinguishing these requires cosmological data beyond the galaxy-scale phenomenology addressed here.

Why Distributed Processing?

Central coordination of gravitational information across cosmic scales would require infinite bookkeeping. Local mediation, where each region processes only its immediate neighborhood, is the architecturally natural alternative (analogous to distributed energy systems in biology). Mass sources distributed throughout space sustain local field gradients. The question then becomes: what is the local capacity? The scale a_0 is where that capacity saturates.

POSTULATE 2: DISTRIBUTED FIELD PROCESSING

The field possesses finite local encoding capacity. Below a characteristic scale a_0, gravitational gradients become indistinguishable from the background. In this regime ($x \ll 1$), the field response becomes scale-invariant:

$$\lim_{x \to 0} \frac{\mu(x)}{x} = 1$$

Derivation of the Asymptotic Form

Given distributed field processing, the linear scaling follows from dimensional consistency. In the deep-field regime ($g \ll a_0$), if the field cannot "see" a_0 as a resolution threshold, then a_0 cannot appear as an explicit scale in the effective dynamics. The only dimensionally consistent combination yielding acceleration from mass that is *independent of a_0 in its functional form* is:

$$g \propto (GMa_0)^{1/2} r^{-1}$$

This implies $g = \sqrt{g_N \cdot a_0}$, where $g_N = GM/r^2$. Comparing with the constitutive relation $\mu(x)g = g_N$, we obtain $\mu(x) = x$ in the limit $x \ll 1$. Thus, linear scaling is the *unique* asymptotic behavior consistent with scale-invariant dynamics under distributed field processing.

POSTULATE 3: SINGLE-CHANNEL CAPACITY SATURATION

The gravitational field operates as a single-channel processor with finite local capacity. At each point, the field response partitions into **resolved** (tracking Newtonian demand) and **unresolved** (saturated) fractions. The equilibrium occupancy $\varepsilon \equiv \mu(x)$ is determined by detailed balance under steady-state processing load.

Although the gravitational potential is static, the constitutive partition is treated as a steady-state balance of mediation load. The transition propensities below are equilibrium throughput weights determining the partition, not time-dependent dynamics.

Definition: Minimal Single-Channel Saturating Closure (MSC)

We define the **MSC closure class** as the minimal parameterization consistent with the physical constraints above:

Axiom	Statement	Physical Meaning	Why Minimal
MSC(i)	Single occupancy variable $\varepsilon \in [0, 1]$	One degree of freedom	Any more adds free parameters
MSC(ii)	Detailed balance: $W_+ = W_-$ at equilibrium	Equilibrium is well-defined	Thermodynamic consistency
MSC(iii)	Linear saturation: $f(\varepsilon) = 1 - \varepsilon$	Capacity-limited response	Leading-order constraint
MSC(iv)	Monomial driving: $g(x) = x^n$, $n > 0$	Power-law response to intensity	EFT leading term
MSC(v)	Constitutive map: $\mu(x) \equiv \varepsilon(x)$	Occupancy = transmissivity	Single-variable ansatz

Note: MSC does not fix the driving exponent n; this is determined by Postulate 2 (see Proposition below). MSC is the baseline closure; departures represent testable extensions. MSC is minimal: it satisfies the empirical asymptotes (Newtonian limit, deep-MOND scaling) with a single internal variable and no free functions.

Interpreting ε. Think of ε as the fill fraction of the closure channel. At $\varepsilon = 0$, the channel is empty (Newtonian regime, high x). At $\varepsilon = 1$, it is full (deep-MOND saturation, low x). At $\varepsilon = 0.5$, the channel is half-filled, which occurs at the processing boundary r_c where $x = 1$. The function $\mu(x) = x/(1 + x)$ describes how the closure channel fills as gravitational intensity decreases.

Why single-channel? If there were two channels, we would need to specify how they interact (another free function). If ε were unbounded, saturation would not be enforced. If the driving were not monomial, we would need additional parameters for the functional form. Each alternative adds complexity. MSC is the minimal viable closure: the simplest parameterization that reproduces the required asymptotes with no free functions.

PROPOSITION 1: DRIVING EXPONENT SELECTION

Within MSC(i–v), Postulate 2 ($\mu/x \to 1$ as $x \to 0$) forces $n = 1$.

Proof sketch: Equilibrium from detailed balance (MSC ii) gives $\varepsilon = Kx^n/(1 + Kx^n)$. With $\mu \equiv \varepsilon$ (MSC v), the small-x limit is $\mu \sim Kx^n$, so $\mu/x \sim Kx^{n-1}$. For this to approach a nonzero constant as $x \to 0$, we require $n = 1$. QED.

*This is a **strength gain**: "linear driving" is now derived within MSC, not assumed. The only inputs are the monomial family (MSC iv) and the empirical deep-MOND limit (P2).*

Methodological note. Proposition 1 uses constraint propagation: given the EFT-minimal monomial ansatz (MSC iv), the empirical constraint (P2) uniquely fixes the exponent. It does not derive the monomial form itself from deeper principles; that family is selected as the leading-order EFT term (analyticity at $x = 0$, no fine-tuned cancellations). The honest claim is this: within the minimal kinetic class, $\mu(x) = x/(1 + x)$ is uniquely determined. Departures from MSC are testable via transition-shape residuals (§3.5).

Extensions beyond MSC. MSC is the baseline closure. Departures correspond to testable signatures:

Extension	Effect	Observable
Multi-occupancy variables	Different $\mu(x)$ shapes	Rotation curve residuals
Nonlinear saturation $f(\varepsilon)$	Different Newtonian approach	High-acceleration deviations
Non-detailed balance	Hysteresis / memory	Path-dependent orbital dynamics
$B \neq L^{-2}$ (A4 relaxed)	Gravitational slip ($\gamma \neq 1$)	Lensing-dynamics mismatch

Extensions introduce additional parameters that must be fit to data. MSC is therefore the baseline; extensions are allowed but measurably constrained by departures from $\mu = x/(1 + x)$.

> **Model Space Disclaimer**
>
> More general closures can produce other $\mu(x)$. We do **not** claim logical inevitability across all closures, only uniqueness inside MSC, which is the minimal closure reproducing the required asymptotes. The uniqueness result (Theorem 1 below) is conditional on the MSC framework.

Why MSC Is the Minimal Consistent Closure

A natural question: why single-channel? Why these specific rate forms? MSC is the minimal closure class consistent with invariances and observed single-scale phenomenology. Each choice has a principled justification:

1. Single channel. Current phenomenology supports one dominant acceleration scale (a_0) with small intrinsic scatter. The SPARC sample (>150 galaxies) and the Radial Acceleration Relation show no evidence for additional transition scales. Multi-channel closures would introduce scale-dependent residuals, testable by comparing AIC/BIC against single-scale models.

2. Stationary reversible closure. The logistic form emerges from minimal two-state kinetics with activation rate $W_+ = \kappa_+ \cdot x \cdot (1 - \varepsilon)$ and deactivation rate $W_- = \kappa_- \cdot \varepsilon$. At equilibrium ($W_+ = W_-$), solving gives $\varepsilon = Kx/(1 + Kx)$. Absorbing the ratio $K \equiv \kappa_+/\kappa_-$ into the definition of a_0 yields $\varepsilon = x/(1 + x)$. This is the unique stationary solution within the minimal two-rate ansatz.

3. Linear driving. The driving variable $x = |\nabla\Phi|/a_0$ follows from a symmetry chain: (i) shift invariance requires dependence on derivatives only; (ii) isotropy requires dependence on the scalar $|\nabla\Phi|$; (iii) analyticity admits a Taylor expansion; and (iv) the deep-MOND limit (P2) forces $n = 1$. Linear driving is uniquely selected within the analytic, isotropic, shift-invariant class.

4. Linear saturation. Finite capacity means a maximum occupancy exists. Expanding the general saturation function $f(\varepsilon)$ near saturation:

$$f(\varepsilon) = c_0(1 - \varepsilon) + c_1(1 - \varepsilon)^2 + \mathcal{O}((1 - \varepsilon)^3)$$

The leading term is linear in available capacity. Linear saturation is therefore the minimal, leading-order capacity constraint.

Connecting to AQUAL: The Constitutive Relation

To make the "resolved fraction" interpretation precise, we define the **Newtonian field** g_N as the field that *would* arise from the same source ρ under standard gravity:

$$\nabla \cdot \mathbf{g}_N = -4\pi G\rho \quad \text{(Poisson equation)}$$

In spherical symmetry, $g_N(r) = GM(r)/r^2$. The AQUAL constitutive relation then reads:

$$g_N = \mu(g/a_0) \cdot g$$

where $g \equiv |\nabla\Phi|$ is the actual (MONDian) field. **Interpretation:** $\mu(x)$ is the fraction of Newtonian demand that the field "resolves." This is interpretive language for the constitutive law, not a hidden extra postulate.

Derivation from Equilibrium Closure

We model the field at each point as existing in one of two states:

- **Resolved (R):** Tracking Newtonian demand faithfully
- **Unresolved (U):** Saturated, producing "phantom" response

Let ε = resolved fraction (equilibrium occupancy). At steady state, the propensity to resolve must balance the propensity to relax:

$$W_+ = \kappa_+ \cdot x \cdot (1 - \varepsilon) \quad \text{(resolution propensity)}$$
$$W_- = \kappa_- \cdot \varepsilon \quad \text{(relaxation propensity)}$$

Physical content. The term $\kappa_+ \cdot x$ represents linear driving by intensity (leading-order closure). The factor $(1 - \varepsilon)$ limits the response by available capacity (one cannot resolve what is already resolved). The term $\kappa_- \cdot \varepsilon$ gives first-order relaxation (resolved states equilibrate with no persistence mechanism).

Corroborating Check: Flat Rotation Curves

The internal consistency argument above is self-contained. As an external check, we verify that $n = 1$ is also uniquely selected by observed galactic phenomenology. Assume spherical symmetry. The constitutive relation $\mu(x)g = g_N = GM/r^2$ with $\mu(x) \approx x^n$ gives:

$$(g/a_0)^n \cdot g = GM/r^2 \quad \Rightarrow \quad g^{n+1} = a_0^n \cdot GM/r^2$$

For circular orbits with $v^2 = gr$, we obtain $v^{2(n+1)} = a_0^n \cdot GM \cdot r^{n-1}$. Flat rotation curves require v independent of r, hence $n - 1 = 0$, giving $n = 1$. Substituting yields the Baryonic Tully-Fisher Relation: $v^4 = a_0 \cdot GM$.

Driving exponent n	Deep-MOND $\mu(x)$	Postulate 2	BTFR form	Rotation curve
1	$\mu \sim x$	✓ Satisfied	$v^4 \propto M$ ✓	Flat ✓
2	$\mu \sim x^2$	✗ $\mu/x \to 0$	$v^{4/3} \propto M^{1/3}$	$v \propto r^{1/6}$ (rising)
1/2	$\mu \sim x^{1/2}$	✗ $\mu/x \to \infty$	$v^{8/3} \propto M^{2/3}$	$v \propto r^{-1/6}$ (falling)

The value $n = 1$ is uniquely selected: it simultaneously satisfies Postulate 2, produces flat rotation curves, and recovers the observed $v^4 \propto M$ Tully-Fisher scaling.

Why force coupling? Why does the driving depend on $x = |\nabla\Phi|/a_0$ rather than x^2 (energy density)? The closure mechanism responds to the demand placed on it, which is the force that needs mediation. Within MSC, Proposition 1 shows that Postulate 2 forces $n = 1$; higher powers yield $\mu \sim x^n$, violating the deep-MOND limit. Both x and x^2 are shift-invariant, but only $n = 1$ satisfies P2. The deep-MOND condition (P2) is an empirical boundary condition, not derived from the kinetics, ensuring no circularity.

At equilibrium (detailed balance): $W_+ = W_-$, giving:

$$\kappa_+ \cdot x \cdot (1 - \varepsilon) = \kappa_- \cdot \varepsilon$$

Solving for ε with $K \equiv \kappa_+/\kappa_-$:

$$Kx(1 - \varepsilon) = \varepsilon \;\Rightarrow\; Kx = \varepsilon(1 + Kx) \;\Rightarrow\; \varepsilon = \frac{Kx}{1 + Kx}$$

Scale absorption (EFT normalization): Since the theory already contains a single empirical scale a_0, any constant K that appears only in the combination Kx can be absorbed by redefining $a_0 \to a_0/K$. This does not introduce a new parameter; it only shifts the fitted value of a_0. Setting $K = 1$ in the rescaled units:

The Gravitational Constitutive Law

$$\mu(x) = \frac{x}{1+x}$$

This is the gravitational constitutive law, recovering MOND's empirical function. What Milgrom discovered empirically now emerges as mathematical necessity.

This is the **unique equilibrium solution within** the single-channel detailed-balance closure class (linear-in-x activation with saturation via $(1-\varepsilon)$, as defined above), not an approximation or truncation.

Stability of the Equilibrium

The equilibrium $\varepsilon^* = x/(1+x)$ is the unique fixed point on $[0,1]$ and is globally attracting. Define the rate of change:

$$\dot{\varepsilon} = W_+ - W_- = \kappa x(1-\varepsilon) - \kappa\varepsilon$$

This is affine in ε with strictly negative slope $d\dot{\varepsilon}/d\varepsilon = -\kappa(x+1) < 0$. At the boundaries: $\dot{\varepsilon}(0) = \kappa x > 0$ (flow pushes away from 0) and $\dot{\varepsilon}(1) = -\kappa < 0$ (flow pushes away from 1). Since $\dot{\varepsilon}$ is strictly decreasing and changes sign exactly once across $[0,1]$, the equilibrium is globally attracting within this kinetic analogy.

Well-posedness. The AQUAL equation derives from a variational principle. For $\mu = x/(1+x)$, the operator is elliptic and monotone. Under standard boundary conditions, existence follows from variational methods, uniqueness from monotonicity, and stability from continuous dependence on data. The DFP boundary-value problem is well-posed under these standard conditions (see Bekenstein and Milgrom 1984).

THEOREM 1: MSC UNIQUENESS

Assume MSC(i–v) and impose:

- **Postulate 1 (P1):** Newtonian recovery: $\mu(x) \to 1$ as $x \to \infty$
- **Postulate 2 (P2):** Deep-MOND scaling: $\mu(x)/x \to 1$ as $x \to 0$

Then the unique equilibrium closure response is:

$$\mu(x) = \frac{x}{1+x}$$

up to rescaling of a_0.

Proof sketch:

1. Equilibrium from detailed balance (MSC ii) gives $\varepsilon = Kx^n/(1+Kx^n)$
2. Proposition 1 (P2 + MSC) forces $n = 1$, yielding $\varepsilon = Kx/(1+Kx)$
3. P1 ($\mu \to 1$ as $x \to \infty$) requires $K > 0$; absorb K into a_0 by rescaling x
4. With $\mu \equiv \varepsilon$ (MSC v): $\mu(x) = x/(1+x)$. QED.

The remaining freedom is the overall scale relating x to $|\nabla\Phi|$, absorbed into the definition of a_0.

> **✓ This Is It**
>
> **μ(x) = x/(1+x)**, *derived*, not fitted, from the architectural constraints of lossless, local, single-channel closure. The function that has been phenomenologically assumed for 40 years emerges uniquely from the structure of field closure.
>
> **What the function means:** At x = 1 (the processing boundary r_c), μ = 1/2, exactly half Newtonian, half MOND. At $x \gg 1$, $\mu \to 1$ (Newtonian). At $x \ll 1$, $\mu \to x$ (deep-MOND). The transition is smooth, monotonic, and uniquely determined.
>
> *This is the function that satisfies the five architectural facts from Section 2. There is no other within MSC.*

Scope of the claim. This theorem does not derive a_0 quantitatively from a capacity bound (it remains an empirical input), nor does it claim this is the only possible physics across all closures. What it does claim: given the observed constraints (P1, P2) and the MSC closure class, $\mu(x) = x/(1+x)$ is uniquely forced. This is a conditional theorem, not a logical inevitability across all model spaces.

(Formal proof: Appendix A, Lemmas A.1–A.4 → Theorem A.5.)

AQUAL action. The constitutive law arises from a standard non-relativistic AQUAL energy functional. Define $X \equiv |\nabla\Phi|^2/a_0^2 = x^2$. In AQUAL, $\mu(x) = dF/dX$, where $F(X)$ is the dimensionless free function, and the Lagrangian density is:

$$\mathcal{L} = -\frac{a_0^2}{8\pi G}\, F(X) - \rho\,\Phi$$

Varying Φ yields the AQUAL field equation $\nabla\cdot(\mu\,\nabla\Phi) = 4\pi G\rho$. Integrating $F'(X) = \sqrt{X}/(1+\sqrt{X})$ gives:

$$F(x^2) = x^2 - 2x + 2\ln(1+x)$$

As $x \to \infty$, $F(x^2) = x^2 + O(x)$, recovering the Newtonian quadratic form; as $x \to 0$, $F \to (2/3)x^3$, the standard AQUAL deep-MOND cubic scaling. This confirms the theory is conservative and variational.

Flow Representation (RG-Like Interpretation)

The derived constitutive law can be equivalently written as the solution to a first-order autonomous flow in the scale-logarithmic variable $\ln x$:

$$\frac{d\mu}{d\ln x} = \beta(\mu) = \mu(1-\mu)$$

Proof. Let $y = \ln x$, so $\mu = e^y/(1+e^y)$. Differentiating with respect to y gives $d\mu/dy = e^y/(1+e^y)^2 = \mu(1-\mu)$. ■

RG-like interpretation (no extra assumptions). This reveals that the MOND transition is an *autonomous* running of an effective coupling μ with "scale" variable $\ln x$ (where $x \equiv |\nabla\Phi|/a_0$). The dynamics has two limits: **UV limit** $\mu \to 1$ as $x \to \infty$ (Newtonian regime) and **IR limit** $\mu \to 0$ as $x \to 0$

(deep-MOND regime). In this sense, MOND phenomenology is not introduced as an ad hoc splice of regimes, but as the **minimal smooth flow connecting two scale-invariant limits**.

This is a *structural analogy* to renormalization-group running; a microphysical coarse-graining derivation is not assumed here. The value of this representation is that it answers "why a special scale?" with: *because flows between fixed points are normal; the scale a_0 is where the crossover happens.*

Why this flow is uniquely determined. The flow coefficient $\mu(1-\mu)$ is the variance of a coin flip with bias μ. Maximum variance occurs at $\mu = 1/2$, exactly where the crossover happens ($x = 1$). The function $\mu = x/(1+x)$ is also the maximum entropy solution: the least biased assignment of constitutive response given that the odds ratio must scale with x. The detailed-balance and entropy-maximization derivations give the same answer.

In the deep-MOND sector ($x \ll 1$), the constitutive response must be scale-free. The only form with this property is a power law for the odds ratio:

$$\frac{\mu}{1-\mu} = C\,x^{k} \quad \Longleftrightarrow \quad \ln\frac{\mu}{1-\mu} = k\ln x + \ln C$$

Postulate 2 fixes the deep-MOND asymptote ($\mu \simeq x$ as $x \to 0$), forcing $k = 1$ and $C = 1$. Hence $\mu(x) = x/(1+x)$. The answer is locked in by symmetry.

Independent Mathematical Check (Padé Uniqueness)

This is a consistency check, not a selection rule: the function derived above happens to coincide with the unique Padé$[1/1]$ form satisfying the same asymptotes.

Within the class of rational functions $P(x)/Q(x)$ satisfying Postulates 1–2, the general Padé$[1/1]$ form is $\mu(x) = ax/(bx+c)$ with three parameters. Three constraints (the two asymptotes plus scale normalization) exactly fix $a = b = c = 1$, yielding $\mu(x) = x/(1+x)$. The equilibrium-closure derivation and Padé uniqueness converge on the same answer from different directions.

Why Alternative Forms Require Additional Physics

To obtain a different $\mu(x)$, one must modify the equilibrium balance. Each modification implies specific additional structure:

Alternative $\mu(x)$	Required Modification	Implies
$x^n/(1+x^n)$ (Hill-type)	$W_+ \propto x^n$	Cooperative/multi-step closure (additional exponent)
$x/\sqrt{1+x^2}$ ("standard")	Add quadratic term	Non-minimal closure (extra curvature coupling)
Non-saturating forms	Remove $(1-\varepsilon)$	Infinite capacity (contradicts finite-resolution postulate)
Irreversible forms	Break detailed balance	Non-equilibrium drive (must specify source)

Burden of proof. Different μ implies additional structure beyond the minimal single-scale, single-channel closure. The burden shifts to the critic to specify what physics produces the deviation.

Universal form. The equilibrium occupancy $y = x/(1+x)$ is the universal solution for systems with: (1) single channel/finite capacity, (2) linear driving, (3) reversible transitions. This mathematical structure appears across physics:

Domain	System	Result
Surface physics	Langmuir isotherm	$\theta = Kp/(1+Kp)$
Biochemistry	Michaelis-Menten	$v/V_{\max} = [S]/(K_m + [S])$
Quantum statistics	Fermi-Dirac ($T \to \infty$)	Same functional form

DFP posits that gravitational mediation has these properties. The function is derived from the equilibrium closure, not borrowed from other fields.

Remark (Reciprocal symmetry and asymptotic rigidity). The derived interpolating function satisfies the exact identity

$$\mu(x) + \mu(1/x) = 1,$$

equivalently $\ln[\mu/(1-\mu)] = \ln x$, a specific property of the logistic odds law $\mu/(1-\mu) = x$. Inversion $x \to 1/x$ swaps $\mu \leftrightarrow 1-\mu$. A key consequence is **asymptotic rigidity**: the deep-MOND limit $\mu(x) \sim x$ for $x \ll 1$ implies the Newtonian recovery $\mu(x) \sim 1 - 1/x$ for $x \gg 1$. (Indeed, from $\mu(x) = 1 - \mu(1/x)$ and $\mu(y) \sim y$ as $y \to 0$, take $y = 1/x$.) Thus, for this closure form the high-acceleration approach is not independently adjustable; quasi-Newtonian tests probe the implied slow $1/x$ recovery.

What Would Falsify This Derivation?

The equilibrium-closure derivation makes specific predictions about the transition shape. Deviations would appear as:

- **Systematic residual structure** in rotation-curve fits near $g \sim a_0$ inconsistent with $\mu = x/(1+x)$
- **Galaxy-dependent scatter** in the RAR beyond measurement uncertainties, implying different systems require different transition parameters
- **Environment-dependence** in the transition behavior after controlling for known systematics (and EFE where applicable)

Current status: Published RAR analyses report intrinsic scatter $\lesssim 0.05$ dex, leaving limited room for additional transition-shape parameters. The single-channel closure is not just theoretically minimal; it is empirically constrained.

3.3 Summary: Three-Layer Derivation

The interpolating function $\mu(x) = x/(1+x)$ is established through three convergent arguments:

1. **Physical mechanism (primary):** The two-state detailed-balance model (Postulate 3) yields $\mu(x) = x/(1+x)$ as the unique equilibrium occupancy of a single-channel, capacity-limited system.
2. **Flow representation:** The same function satisfies $d\mu/d\ln x = \mu(1-\mu)$, the minimal beta-function flow connecting the deep-MOND and Newtonian fixed points.
3. **Padé uniqueness:** Within the class of rational interpolants satisfying Postulates 1-2, $\mu(x) = x/(1+x)$ is the unique Padé[1/1] form. Three constraints exactly fix three parameters.

Convergence: Physical mechanism, formal flow, and mathematical uniqueness all yield the same function. This is not coincidental; it reflects that DFP's single-channel, single-scale architecture admits only one consistent constitutive law.

Attack surface: To argue for a different $\mu(x)$, a critic must identify which term in the equilibrium balance is wrong and specify the replacement mechanism. The burden of proof shifts from "why this function?" to "what additional physics produces the deviation?"

3.4 The Field Equation

With the constitutive law derived, we now embed it in the field equation that governs dynamics. The equation is governed by the AQUAL action:

$$\mathcal{S}[\Phi] = \int d^3x \left[\frac{a_0^2}{8\pi G} \mathcal{F}(X) - \rho(\mathbf{x})\Phi(\mathbf{x}) \right]$$

where:

$$X \equiv \frac{|\nabla\Phi|^2}{a_0^2}, \quad \mu(x) = \frac{d\mathcal{F}}{dX}\Big|_{X=x^2}$$

(Note: $x \equiv |\nabla\Phi|/a_0 = \sqrt{X}$, so μ is evaluated at $X = x^2$.)

Varying with respect to Φ yields the modified Poisson equation:

$$\nabla \cdot \left(\mu\left(\frac{|\nabla\Phi|}{a_0} \right) \nabla\Phi \right) = 4\pi G\rho$$

3.5 Relation to Existing MOND Interpolating Functions

The function $\mu(x) = x/(1+x)$ derived above is not new. It is the "simple" interpolating function introduced by Famaey & Binney (2005) and widely used in MOND phenomenology. Our contribution is not the function itself, but the **derivation from physical mechanism** (equilibrium closure of a single-channel capacity-limited system) that makes it the unique solution rather than a fitting choice.

The MOND literature contains several interpolating functions, all satisfying the same asymptotic limits:

Name	Form	Rational Class	Undetermined Parameters
Simple (DFP-derived)	$\mu = x/(1+x)$	Padé [1/1]	**0** (derived from mechanism)
Standard	$\mu = x/\sqrt{1+x^2}$	Padé [1/2]	1 (exponent not fixed by closure)
RAR-fit	$\mu = 1/(1+e^{-x})$ variants	Transcendental	∞ (no closure constraint)

Empirically, the simple form is favored. Gentile et al. (2011) found that it yields stellar mass-to-light ratios consistent with population synthesis models, whereas the standard form often requires unphysical values. The simple form also provides statistically superior fits for high surface brightness galaxies in the THINGS sample.

The DFP contribution: Previous work selected $\mu(x) = x/(1+x)$ because it fits data well. We show it is the *unique equilibrium solution* of a single-channel capacity-limited system under detailed balance, combined with Newtonian correspondence and finite encoding capacity. The function is not merely empirically preferred; it is **implied by the closure assumptions** within that class.

3.6 The Claim Ledger

The following table tracks the logical status of each claim: what is empirical input, what is a structural axiom, what is a closure-class assumption, and what is derived.

Claim	Status	Depends On	Where
Deep-MOND scaling ($\mu/x \to 1$)	Empirical (P2)	Observations	§3.2
Newtonian recovery ($\mu \to 1$)	Empirical (P1)	Observations	§3.2
No quasi-static hysteresis	Empirical	Observations	§2.4
Single occupancy variable	MSC(i)	Minimal ansatz	§3.2
Detailed balance	MSC(ii)	Minimal ansatz	§3.2
Linear saturation	MSC(iii)	Minimal ansatz	§3.2
Monomial driving family	MSC(iv)	Minimal ansatz	§3.2
Constitutive map $\mu \equiv \varepsilon$	MSC(v)	Single-variable ansatz	§3.2
Linear driving ($n = 1$)	**Derived**	P2 + MSC(i-v)	§3.3
$\mu(x) = x/(1+x)$	**Derived**	P1, P2, MSC(i-v)	§3.3
$\gamma = 1$ (no gravitational slip)	Corollary of A4	A4 (operational)	§7.2

Key: "MSC" = Minimal Single-Channel Saturating Closure (defined in §3.2). "P1/P2" = Postulates 1 and 2 (empirical boundary conditions). "A4" = Calibration Postulate (operational definition of rod-clock calibration, §7.2).

The derived results (highlighted rows) follow necessarily from the inputs. The attack surface is the MSC closure class itself: to obtain a different $\mu(x)$, one must specify what additional physics produces the deviation from minimal single-channel closure.

4. Consequences for Galaxies

Section 3 derived the constitutive law $\mu(x) = x/(1+x)$. This section tests it: does this law reproduce the observations that motivated the entire investigation? We show it delivers the Tully-Fisher relation, explains "dark matter halos," and predicts environmental effects absent in particle dark matter.

4.1 The Baryonic Tully–Fisher Relation

In §1.1, we noted the remarkably tight Baryonic Tully-Fisher Relation: $M_b \propto v^4$ with minimal scatter. Within DFP, this is not a correlation to be explained; it is a theorem of the constitutive law. When gravitational fields are weak ($x \ll 1$), the constitutive law simplifies: $\mu(x) \approx x$. For an isolated system of mass M_b:

$$\mu(x) g = \frac{G M_b}{r^2} \implies \frac{g^2}{a_0} \approx \frac{G M_b}{r^2}$$

(Exact in spherical symmetry; more generally valid when the curl field is negligible; the full field equation is AQUAL.)

Solving for the circular velocity $v_c^2/r = g$, we recover the **Baryonic Tully–Fisher Relation**:

$$v_c^4 = G M_b a_0$$

This is an **asymptotic theorem**, not a fit. The exponent 4 and the coefficient $G a_0$ are derived, not assumed. The BTFR emerges from the constitutive law; it is not imposed upon it.

The BTFR is a theorem. But what about the "dark matter halos" inferred from rotation curves?

4.2 Origin of the Phantom Density

The Introduction asked: why does "missing mass" correlate so precisely with visible matter? Here is the answer. If there is no dark matter, what does a Newtonian observer infer? The answer is precise and calculable. Standard linear gravity infers a total density ρ_{eff} via $\nabla \cdot \mathbf{g} = -4\pi G \rho_{\text{eff}}$. The phantom density (i.e., the density inferred by Newtonian inversion, not a physical component) is $\rho_{\text{ph}} = \rho_{\text{eff}} - \rho_b$. Throughout, $\mathbf{g} \equiv -\nabla\Phi$ denotes the gravitational field (with magnitude g and $x \equiv g/a_0$), and we use the sign convention $\nabla \cdot \mathbf{g} = -4\pi G \rho_{\text{eff}}$.

What exactly is this "missing mass"? We define it precisely. Let $\mathcal{L}_{\text{N}}[\Phi] \equiv \nabla^2\Phi$ be the Newtonian (linear) operator and $\mathcal{L}_{\text{DFP}}[\Phi] \equiv \nabla \cdot (\mu \nabla \Phi)$ be the DFP (nonlinear) operator. The phantom density is the **operator residual**:

$$\rho_{\rm ph} \equiv \frac{1}{4\pi G}\left(\nabla^2\Phi - \nabla\cdot(\mu\nabla\Phi)\right) = \frac{1}{4\pi G}\nabla\cdot[(\mu-1)\mathbf{g}]$$

Here Φ is the DFP/AQUAL potential solving $\nabla\cdot(\mu\nabla\Phi) = 4\pi G\rho_b$. The "phantom" density is what a Newtonian inversion would infer if it applied ∇^2 to that same Φ. The phantom density is nonzero precisely because $\mu \neq 1$: it measures the discrepancy between nonlinear (resolution-limited) and linear (resolution-unlimited) inversions of the same field.

For $r > 0$ (outside the baryonic support), in the deep-MOND regime this yields:

$$\rho_{\rm ph}(r) = \frac{\sqrt{GM_b a_0}}{4\pi G r^2} \propto r^{-2}$$

This is exactly the **singular isothermal sphere** profile, derived rather than assumed. The "dark matter halo" is an analytic artifact of applying linear tools to a nonlinear system.

The Static Nature of the Transition

A common misconception: that MOND "amplifies" weak signals. The truth is simpler. The constitutive factor μ depends on $|\nabla\Phi|$ (the field gradient), not on velocity. The modification is therefore *static*: it describes the shape of the equilibrium solution, not a dynamic response to motion.

In spherical symmetry, the AQUAL equation implies a conserved flux:

$$r^2\mu(g/a_0)\,g = GM$$

This is the μ-weighted flux constraint. The "extra gravity" at large r is a static geometric consequence of this constraint, not an amplification of a fading signal. When $\mu(x) \approx x$ for $x \ll 1$, the deep-MOND limit follows immediately:

$$(g/a_0) \cdot g = GM/r^2 \quad \Rightarrow \quad g = \sqrt{GMa_0}/r$$

There is no "fading signal" that requires "amplification." The field simply satisfies a different constitutive law in the weak-field regime. The phantom density is the operator residual: what you get when you apply the Newtonian operator to a non-Newtonian field.

The baryon-halo "conspiracy" becomes a non-mystery. Within DFP, the "phantom" component is an operator residual of the baryonic potential under the nonlinear constitutive law, not an independent mass component whose correlation with baryons must be explained separately. Phantom density explains halos. But one prediction distinguishes DFP from CDM entirely.

4.3 Sensitivity to External Fields

AQUAL-class theories predict something dark matter cannot: internal dynamics that depend on external fields. This is the **External Field Effect (EFE)**. The nonlinearity of the field equation implies that the internal dynamics of a subsystem depend on external gravitational fields. This is not a new prediction of DFP; it is inherited directly from AQUAL's mathematical structure.

> **EFE in DFP Terms: The NET Field Connection**
>
> In DFP terminology, the external field is the **NET Field**: the environmental configuration in which the field object is embedded. The EFE is what DFP predicts: the NET Field affects internal dynamics because closure happens at the interface, not in isolation. At Σ, the combined gradient determines μ(x); when the NET Field dominates, the subsystem is "lifted" into the Newtonian regime.

Consider a subsystem (e.g., a dwarf galaxy or star cluster) with internal Newtonian acceleration $g_{\rm int}$ embedded in a host potential with external field $\mathbf{g}_{\rm ext}$. The total gradient is $|\nabla\Phi| = |\mathbf{g}_{\rm int} + \mathbf{g}_{\rm ext}|$. In the limit where $|\mathbf{g}_{\rm ext}| \gg a_0$ but $|\mathbf{g}_{\rm int}| \ll a_0$:

$$\mu\left(\frac{|\mathbf{g}_{\rm int} + \mathbf{g}_{\rm ext}|}{a_0}\right) \approx \mu\left(\frac{g_{\rm ext}}{a_0}\right) \approx 1$$

The subsystem then obeys *Newtonian* dynamics despite its internal accelerations being sub-threshold. The external field "lifts" the local gradient above the resolution floor. In DFP terms: the external field provides additional gradient structure that the field *can* resolve. The information

content of $\nabla\Phi$ is dominated by the external contribution, keeping internal dynamics near-Newtonian by driving $\mu \approx 1$. The external field "uses up" the nonlinear regime, leaving no room for the resolution-limited response to manifest.

EFE Prediction: Satellite galaxies in the inner regions of massive hosts (where $g_{\rm ext} \gtrsim a_0$) should exhibit reduced mass discrepancies compared to isolated dwarfs of similar baryonic mass. This effect has been tentatively observed in Crater II and other satellites, and provides a critical test distinguishing DFP from CDM, where no such environmental dependence exists. This environmental dependence provides the critical test promised in the Abstract.

Why this matters: The EFE is arguably MOND's most distinctive prediction, and its most sensitive test. Wide binary studies and satellite galaxy dynamics probe exactly this regime. DFP inherits this prediction without modification; the physical interpretation (encoding capacity saturation) aligns naturally with the mathematical structure.

With phenomenology established, Section 5 examines the mechanism: what physical principle produces this μ(x), and why does a_0 take the value it does?

5. Field Closure

Section 3 derived μ(x). Section 4 showed its consequences. Now we address the underlying physics: *how do fields couple?*

When our asteroid drifts through the Solar System, it doesn't just "feel" the Sun's gravity as an external force. The asteroid's field configuration and the gravitational environment must become *compatible*. They must couple into one consistent state. This coupling doesn't happen by magic. Something must read both configurations, enforce compatibility, and produce a single registered result.

That something is field closure. This section explains where it happens (Σ), what it does (FC_Σ), and why it must be lossless.

5.1 The Coupling Problem

The problem: A field object and the NET Field are both quantum configurations. When they meet at Σ, they exist in the same local conditions. They don't simply coexist; they must *couple* into a consistent state. The asteroid's 10^{33} fields and the Sun's gravitational field must resolve to one compatible configuration. What mechanism accomplishes this?

In modern physics, matter is quantum field state. The asteroid is not a collection of classical particles; it is a unified quantum many-body configuration. The gravitational environment (the NET Field) is also quantum field state. Both are configurations that must interface. The answer lies in

their shared locality: because they occupy the same local space at Σ, compatibility is ensured by co-location.

Field closure (FC_Σ) is the mechanism that couples them. At the closure boundary Σ, both configurations share the same local field state. FC_Σ reads both configurations in full, enforces compatibility, and settles them into a unique registered state. No global coordination is needed because co-location guarantees they can couple. Here is the complete chain:

> **The Canonical Chain: From Inputs to Registration**
>
> Field Object + NET Field
>
> (inputs)
>
> ↓
>
> **Closure Boundary (Σ)**
>
> (where)
>
> ↓
>
> **FC_Σ: Field Closure Function**
>
> READ → SETTLE
>
> ↓
>
> **B^*_Σ: Registration**
>
> (settlement complete)
>
> ***The standard sentence:*** *"The field closure function FC_Σ operates at the closure boundary Σ: both configurations share the same local geometry, enabling a lossless read, and the two fields settle into a consistent state. The output is the registered boundary state B^*_Σ; gradients $\nabla\ell(x)$ are gravity."*

Definition (Field Closure Function). The Field Closure Function $\mathbf{FC_\Sigma}$ (or simply **field closure**) is the function that governs how the NET Field interfaces with a field object. It takes two inputs (field object configuration, NET Field configuration) and produces one output (the registered boundary state B^*_Σ). The function operates in two steps:

- **READ:** Full-State Snapshot (FSS) reads both configurations in full, lossless because both share the same local geometry $\ell(x)$
- **SETTLE:** Configurations couple to a consistent state; gradients $\nabla\ell(x)$ = gravity

FC_Σ is not a physical field or particle; it is the function that enforces compatibility at Σ while accounting for the entire field object. Any well-posed boundary-value problem requires such a closure function.

The exterior field exhibits a derived threshold contour $S_1 \equiv \{x = 1\}$, which in spherical symmetry occurs at radius $r_c = \sqrt{GM/a_0}$.

> **Σ vs. r_c: The Stage vs. A Specific Choice**
>
> **Closure boundary (Σ)** is the geometric locus where the constraint is enforced, it's the *stage*. **r_c** (the processing boundary where x = 1) is one *specific choice* of Σ commonly used in spherically symmetric applications.
>
> Σ is *input* (the locus where we enforce the constraint). FC_Σ produces B^*_Σ (the registered boundary state) as *output*. In other words: **Σ is the stage; B^*_Σ is the signed contract.**
>
> *$S_1 \equiv \{x = 1\}$ and $r_c = \sqrt{GM/a_0}$ are derived contours with no independent degrees of freedom; they emerge from the field equations, they are not additional inputs.*

The Chain in Action. Our asteroid (the field object) drifts through the Solar System (the NET Field from Sun, planets, distant stars). At the closure boundary Σ, both configurations exist in the same local conditions. Field closure enforces compatibility, recognizing that their co-location ensures they can settle into a consistent state. The result: a gravitational field that other objects respond to. This is the canonical chain, instantiated, all happening locally, without global coordination.

The novel physics. Field theory already requires closure; any gravitating source has a boundary where coupling occurs. The novel contribution is specifying what closure *does*: when the field object and NET Field settle together at Σ, where they share the same local conditions, the constitutive response μ(x) saturates at finite capacity, producing the resolution floor at a_0.

POSTULATE (LOSSLESS CLOSURE)

The field closure boundary Σ supports a finite-capacity boundary state space $\mathcal{H}_\Sigma$ and an encoding map from the domain's physical state (within the relevant code subspace) into $\mathcal{H}_\Sigma$. In this sense, field closure provides **complete encoding** of the coupled region: the boundary state is a complete representation of the field object's state, not a lossy summary.

Causality. This postulate does not introduce superluminal influence. Excitations propagate outward at *c* through the field dynamics; lossless encoding ensures that outward propagation does not sever the domain's state consistency. No-signaling is preserved: boundary correlations constrain joint statistics but cannot be used to transmit controllable messages.

For readers familiar with effective field theory, the following table translates DFP terminology into standard EFT language:

DFP Term	EFT Translation
Processing	Boundary state update + coupling to field solution
Field closure (FC_Σ)	Boundary condition enforcing consistency at Σ; operates via READ → SETTLE
Lossless closure	Complete encoding within a code subspace
Closure boundary (Σ)	Boundary locus where constraint is enforced (geometric input)
Capacity saturation	Breakdown of linear response below a_0
S_1 / r_c	Derived threshold contour {x=1}; no independent DOF

This section and the next will develop each step of the chain. Here, we explain *where* closure happens (the closure boundary Σ), *what* the field closure function is (FC_Σ), and *why* it must be lossless. Section 6 will complete the picture: the process (settlement), the result (registration), and the quantification (r_c).

5.2 The Closure Boundary (Σ): Where Fields Interface

The closure boundary Σ is the geometric locus where compatibility is enforced. At Σ, both configurations exist in the same local conditions; they share the same local field state. This is the *stage* on which the field closure function operates. Think of Σ as "where" and field closure as "what happens there."

The Asteroid's Closure Boundary. Where is the asteroid's Σ? At the geometric locus where the asteroid's field object and the NET Field exist together in the same local conditions. At Σ, both configurations share the same local field state, and this shared locality ensures they can couple. Σ is not a physical surface you can touch; it is a role in the field equations. The closure boundary is input (we identify where both configurations are co-located); the registered state B^*_Σ is output.

5.3 The Field Closure Function (FC_Σ)

FC_Σ is the function that couples the field object and NET Field at the closure boundary Σ. At Σ, both configurations share the same local conditions. FC_Σ enforces compatibility between them and produces the registered boundary state B^*_Σ.

Recall the chain: READ → SETTLE. Both configurations share the same local geometry $\ell(x)$, enabling a lossless read; settlement resolves them into one consistent state. The output B^*_Σ is the gravitational field that distant observers respond to; gradients $\nabla\ell(x)$ are gravity.

FC_Σ operates continuously. Because coupling is continuous (Section 2.2), so is field closure. As the asteroid moves through the NET Field, FC_Σ doesn't fire once; it operates continuously, producing a continuously updated registered state. The gravitational field doesn't lag behind the asteroid; it follows because FC_Σ is continuously complete.

Information propagates outward. When the asteroid couples with the NET Field, its configuration is encoded at Σ and propagates outward into the gravitational field. The field carries the asteroid's signature; that's why distant observers feel its gravity. This is encoding, not trapping. (Unlike an event horizon, where information falls inward, field closure sends information outward into the observable field.)

> **Running Example: The Asteroid's Signature**
>
> When our asteroid couples to the NET Field, its configuration is encoded at Σ and propagates outward. The gravitational field carries the asteroid's signature. That's why distant observers feel its gravity, why spacecraft trajectories bend around it, why it contributes to the NET Field that other objects experience.

NECESSITY CLAIM (FIELD CLOSURE FC_Σ REQUIRED)

Given the DFP postulates in Section 2, field closure (FC_Σ) is required at the boundary Σ. We do not assume field closure is a physical object; we assume only the functional role needed for well-posed closure. The derivation does not proceed without it.

5.4 Why Lossless: The Distributed Forgetting Test

Why must closure be lossless? Because if it weren't, we would observe the consequences. The name for the failure mode is *distributed forgetting*, and naming it matters because it turns a vague worry into observable signatures.

Distributed Forgetting. If snapshot accounting were even *slightly* lossy at each interaction, if some correlations were not registered and instead discarded as "too detailed," then the theory would quietly become a dissipative, history-dependent medium. That is exactly the failure mode this framework forbids, because gravity arises from high-fidelity accounting, not from a medium that leaks information.

Why physics requires losslessness. Hawking's resolution of the black hole information paradox established that gravitational systems must preserve information; erasure violates unitarity. This is physics, not philosophy. If the field forgot at every boundary, complexity would hit a ceiling: structure builds, but nothing persists past the boundary. This is the distributed forgetting that unitarity forbids.

Why saturation implies closure. A field that processed without saturating would require infinite capacity at every point, excluded by the Bekenstein bound. Locality plus finite capacity implies saturation boundaries. Field closure is not imported from outside; it is where DFP's own assumptions force a boundary to appear.

If Closure Were Lossy. A lossy closure update cannot hide; it must show up as at least one of:

- **Effective drag / wake:** unregistered correlations shed, motion leaves a trace
- **Hysteresis:** field response depends on path taken, not instantaneous configuration
- **Spurious heating:** discarded correlations appear as entropy production
- **Apparent nonconservation:** lossy updates leak energy-momentum

We don't observe any of these. Asteroids drift through the Solar System without leaving gravitational wakes. The field is static, elliptic, conservative. **The distributed forgetting test passes.**

By naming the failure mode, we make "FSS must be lossless" a testable constraint, not an interpretive preference. FSS prevents the framework from collapsing into an empirically ruled-out category: draggy, hysteretic, dissipative modified gravity.

5.5 The Quantum Nature of Field Closure

At the closure boundary Σ, both configurations are quantum field states sharing the same local conditions. The field object (10^{33} atoms in correlated field state) and the NET Field (the gravitational environment) are not classical objects. They are quantum many-body configurations. Field closure is where they couple.

Both inputs are quantum. Matter is quantum field state. Our asteroid is not a classical object that happens to obey quantum mechanics at small scales. It is a quantum many-body configuration held together by electromagnetic binding. The NET Field is also quantum field state. This is not speculation; this is quantum field theory.

Co-location enables coupling. At Σ, the field object and NET Field share the same local field state. Their co-location ensures compatibility: they can couple because they exist in the same local conditions. No global coordination is needed. The coupling is local, automatic, and complete.

The quantum-to-classical transition. Two quantum field states (object and environment) couple via field closure at Σ. Settlement enforces compatibility. The output is a settled gravitational potential Φ*, the classical field that test particles follow, that light bends around, that clocks slow within. This is where quantum meets classical: the registered state B^*_Σ is the gravitational field we observe.

Why this isn't superluminal. A common worry: if field closure involves quantum correlations across the closure boundary, doesn't that imply faster-than-light signaling? No. The mechanism is local. At Σ, both configurations share the same local conditions; coupling happens there, not across space. The elliptic field equation relates the gravitational field to the matter distribution as an instantaneous constraint on a time-slice, not as a signaling process. This is exactly analogous to the Coulomb potential in Coulomb gauge: correlations without communication.

Does "field closure" mean measurement? No. Field closure is a consistency constraint at Σ, not a measurement or classical recording. No observer, no log entry. The closure is algebraic: two configurations sharing the same local conditions settle into a consistent state.

Does "complete encoding" mean infinite memory? No. The boundary state space H_Σ has finite capacity. "Complete" means lossless representation within this finite capacity, not infinite storage. This finite capacity is precisely what produces saturation at a_0. The constitutive response μ(x) saturates because the local field state has finite capacity for encoding.

For readers who want the field-theory framing: this is standard EFT practice. Specify the action, the field content, the boundary data. Leave the UV completion underdetermined. All quantitative predictions (rotation curves, bar pattern speeds, Bullet Cluster constraints) follow from the AQUAL field equation with μ(x) and its transition across x = 1. They do not depend on which microphysical interpretation is correct.

6. Settlement and Registration

Section 5 defined the stage (Σ) and the field closure function (FC_Σ). Now we complete the picture: what FC_Σ *does* (settlement), what the result is (registration), and how we quantify it (r_c).

6.1 Settlement: The Process

Settlement is what field closure *does*. When the field object couples with the NET Field at the closure boundary Σ, where both exist in the same local conditions, field closure enforces compatibility. Because both configurations share the same local field state, they are inherently compatible.

Settlement is the process by which this coupling resolves to a unique state through their shared local conditions.

How the Asteroid Settles. When our asteroid enters the Solar System, its quantum field configuration couples with the NET Field (the combined gravitational environment from Sun, planets, galaxy). At Σ, both configurations exist in the same local conditions. Field closure at Σ enforces compatibility, recognizing that their co-location ensures they can settle into a consistent state. This is not computation; it is constraint satisfaction made automatic by co-location.

Settlement is not a separate mechanism from field closure; it is what the field closure function *does*. The terminology distinguishes the function (FC_Σ) from its execution (settlement). Processing is the method: distributed constraint enforcement via the elliptic field equation.

6.2 Field State Registration: The Result

Field state registration (B^*_Σ) is the outcome of settlement. It is the settled boundary state: what the field looks like after closure has enforced compatibility. This is the "signed contract" from the canonical chain.

> **Running Example: The Asteroid's Full Configuration**
>
> Our asteroid isn't a list of 10^{33} particles with positions. It's a unified chord, a correlated many-body field state where the binding relationships are as real as the atoms themselves. When field closure registers the asteroid, it registers the *whole configuration*: the nickel-iron crystalline structure, the internal stresses, the binding energies. Snapshot accounting means full-state snapshot, not surface sampling.

> **Canonical Definition: Full-State Snapshot (FSS)**
>
> FSS is defined operationally: the closure update introduces **no effective dissipation, no hysteresis, and no wake/drag** in the static limit. This is testable: if we observed gravitational wakes from moving asteroids, FSS would be falsified. We don't observe them.
>
> Full-state snapshot means the boundary state is a complete representation of the domain's state within the physical code subspace, not a lossy summary.

The Standard Chain + Expanded View

```
Field Object + NET Field
        ↓
Closure Boundary (Σ)  ← WHERE (same local geometry ℓ(x))
        ↓
FC_Σ: Field Closure Function
    ├── READ: Full-State Snapshot (FSS)
    │         (lossless—same local geometry ℓ(x))
    └── SETTLE: Configurations couple → ∇ℓ(x) = gravity
                (detailed balance, reversible, no information loss)
        ↓
B*Σ: Registration  ← THE RESULT
```

What happens inside field closure? The field object configuration and the NET Field configuration must couple to a consistent state. Two things happen: **Read** (both configurations are captured via Full-State Snapshot, lossless because they share the same local geometry $\ell(x)$), and **Settle** (configurations couple to a consistent state; $\nabla\ell(x)$ = gravity).

Coupling Load. Coherent coupling of a quantum configuration scales with its coupling load. The read is lossless because both configurations share the same local geometry $\ell(x)$. Once read, configurations couple to a consistent state. A larger configuration has more to couple: a moon's field chord is larger than an asteroid's, so more gravity is coupled across that configuration. This is why larger configurations have more mass. $\ell(x)$ specifies local clock rates and length scales; gradients $\nabla\ell(x)$ are gravity.

The key insight: Field closure couples the field object configuration with the NET Field. This coupling works because both configurations exist in the same local conditions at Σ; they share the same local field state. This coupling is continuous: as fields interact, field closure maintains consistency between them through their shared local conditions. Gravity is applied uniformly across the exact configuration, not as a spherical average, but respecting the actual structure. The coupling follows the configuration: where the field object is, gravity couples with it, because both exist in the same local space.

Coupling load is not an independent charge on top of mass-energy; it IS the operational measure of effective mass-energy.

The local geometry scale $\ell(x)$ describes the metric at each position—including clock rates. Gradients $\nabla\ell(x)$ are what we experience as gravity.

6.3 The Processing Boundary (r_c): Quantifying Closure

The processing boundary r_c is where saturation occurs: where $\mu(1) = 1/2$ and the field transitions from Newtonian to deep-MOND. For the asteroid, r_c is microscopic. For the Milky Way, $r_c \approx 8$ kpc, near the solar circle.

The processing boundary radius is defined as the surface where gravitational acceleration equals the resolution scale:

$$g(r_c) = a_0, \quad \text{equivalently} \quad x(r_c) = 1$$

For a spherically symmetric mass distribution with total baryonic mass M_b:

$$r_c = \sqrt{\frac{GM_b}{a_0}}$$

Numerical example: For a Milky Way-like galaxy with $M_b \approx 6 \times 10^{10}\, M_\odot$:

$$r_c \approx 8\,\text{kpc}$$

This places the processing boundary near the solar circle, consistent with where the Milky Way's rotation curve transitions from rising to flat.

The Asteroid's r_c. For our 33-billion-tonne asteroid, $r_c = \sqrt{GM/a_0} \approx 4{,}300$ km, far larger than the 10 km asteroid itself. The surface sits deep within the Newtonian regime ($x \gg 1$), so the near-field is entirely classical. Closure exists, but MOND phenomenology only emerges at distances approaching r_c.

Coherent vs. Composite Sources. The formula $r_c = \sqrt{GM/a_0}$ applies to gravitationally coherent sources: compact mass distributions whose potential is dominated by a single center. For composite systems like globular clusters ($N \sim 10^5$ stellar sources), the cluster potential is a superposition:

$$\Phi_{\text{cluster}}(\mathbf{r}) = \sum_{i=1}^{N} \Phi_i(\mathbf{r} - \mathbf{r}_i)$$

6.4 Universality: Every Source, Every Scale

Field closure is not restricted to stars and galactic centers. Every gravitating source has its own closure: the asteroid, the Sun, the galaxy. The difference isn't whether closure exists; it's whether r_c falls at observable radii.

The Asteroid's Perspective. Our asteroid doesn't borrow closure from the Sun. It has its own closure boundary Σ, its own field object (10^{33} atoms of nickel-iron), and its own registration. At Σ, the asteroid's field object and the NET Field (which includes the Sun's contribution) exist together in the same local conditions; they share the same local field state. The Sun's gravity is part of the NET Field present locally. This is *encapsulation*: each source closes its own field locally, where its field object and the NET Field share the same local conditions.

> **The Fundamental Rule**
>
> Wherever gravity is non-zero ($g \neq 0$), the source's field satisfies a local closure constraint. This is not an active computation performed by matter; it is the self-consistency requirement that the AQUAL field equation imposes on any mass distribution. The field "accounts" for itself transactionally within its own domain. No handoffs, no custody transfers, no cross-scale micromanagement.

Scale determines observability, not existence. While closure is universal, its observable consequences depend on where r_c falls:

- **Planets and asteroids:** r_c is microscopic or lies within the object. The exterior field is entirely Newtonian ($x \gg 1$). Closure is satisfied, but no MOND phenomenology is observable.
- **Stars:** r_c extends to AU scales. The transition $x = 1$ occurs at observable radii, but still far from typical observational contexts.
- **Galaxies:** r_c extends to kpc scales. The saturation regime ($x \ll 1$) dominates where rotation curves and lensing are measured. Phantom structure becomes prominent.

The physics is identical everywhere. The difference is scale.

7. Testable Predictions

Sections 3-4 derived $\mu(x)$ and showed it reproduces galactic phenomenology. Sections 5-6 explained why this derivation is valid. But a theory that merely reinterprets observations is not science; it must make predictions that could prove it wrong. This section presents three: one that particle dark matter cannot match (fast bars), one that establishes when AQUAL applies (quasi-static validity), and one that tests the specific functional form derived in Section 3 (wide binaries).

7.1 Fast Galactic Bars

The Introduction noted that if "missing mass" were an independent substance, its correlation with baryons would be mysterious. Here we show the converse: because the phantom halo is not an independent substance, it cannot generate the dynamical friction that particle halos produce. This distinction yields DFP's most distinctive prediction.

The CDM Problem

In particle dark matter models, galactic bars must plow through a sea of collisionless particles. This generates wakes that transfer angular momentum from bar to halo, slowing the bar's pattern speed over time. Simulations must tune halo concentrations, core profiles, and feedback mechanisms to minimize this friction and explain observed fast bars. CDM fights against its own physics.

The DFP Solution

In DFP, the phantom halo is not a particle reservoir but an operator residual: the discrepancy between nonlinear field dynamics and linear Newtonian inference. Three features of AQUAL preclude halo-wake dynamical friction in the quasi-static limit:

> **Three Features Precluding Friction**
>
> 1. **Conservative dynamics:** AQUAL derives from an action principle; energy, momentum, and angular momentum are conserved.
> 2. **No velocity dependence:** The constitutive factor μ depends on $|\nabla\Phi|$ (field gradient), not on velocity.
> 3. **Phantom is constraint-slaved:** The phantom component is not an independent reservoir; it tracks the baryonic potential instantaneously.

The key intuition: A particle halo is like a boat moving through water; the water resists (drag). A phantom halo is like a spotlight on a wall; the wall does not resist the light. There is no medium to create friction. Velocity does not appear in the AQUAL field equation; it never did. The quasi-static framework is elliptic, with no time-retarded response that could generate drag.

$$\mathcal{R}_{\rm DFP} \approx 1.0\text{–}1.4 \quad \text{(fast bars)}$$

This contrasts with $\mathcal{R}_{\rm CDM} \gtrsim 1.4\text{–}2.0$ (slow bars) after several Gyr of dynamical friction. Fast bars are a high-leverage discriminant: particle-halo frameworks generically predict bar slow-down unless halo structure conspires to suppress it.

Observational status: The preponderance of fast bars in the local universe, including the Milky Way bar, is consistent with DFP predictions and challenges halo-friction models. Internal baryonic redistribution and gas dissipation remain possible friction channels, but these are distinct from halo-wake braking. This is the first falsifiable prediction promised in the Abstract: fast galactic bars.

Fast bars test whether "dark matter" is an independent substance. But to interpret any observation, we must first understand when AQUAL applies.

7.2 Quasi-Static Validity and Transients

AQUAL is an elliptic theory: no time derivatives, no response delays, no propagation speed for gravitational changes. What does this mean for real astrophysical systems? It defines when the framework applies and how to interpret transient phenomena.

The Elliptic Nature

The non-relativistic sector of DFP takes the AQUAL form: an elliptic boundary-value theory. Solutions are determined by the snapshot configuration $\rho(\mathbf{x})$ at a given time and the boundary conditions. There is no gravitational "response time" in the present framework. This is analogous to electrostatics being the static limit of Maxwell's equations: the constraint sector of an underlying causal theory.

> **Scope: Quasi-Static Regime**
>
> The quasi-static treatment applies when source evolution is slow enough that retardation effects are negligible. For equilibrated galaxies where $\partial\rho/\partial t \approx 0$ on dynamical timescales, AQUAL applies directly. Time-dependent phenomena require the covariant completion (Section 8.2).

Transient Systems

For rapidly evolving configurations (major mergers, starbursts), the quasi-static framework treats the system as a family of snapshots, each satisfying closure continuously. Observable timescales in cluster mergers (gas sloshing, shock propagation, ram-pressure stripping) arise from baryon dynamics (hydrodynamics, collision geometry, gravitational free-fall), not from the constitutive law $\mu(x)$.

The Bullet Cluster offset (Section 8.5) is explained by spatial structure: compact galaxies produce sharper phantom peaks than diffuse gas. The field equation is elliptic; there is no time-dependent "lag" or "response time" to invoke. Merging clusters are stress-tests of the instantaneous field-to-lensing mapping, not probes of gravitational propagation.

Within the quasi-static regime, the derived $\mu(x)$ makes one more prediction: the specific shape of the Newtonian transition.

7.3 Wide-Binary Discriminant

The derived $\mu(x) = x/(1+x)$ makes a specific prediction about the return to Newtonian behavior. Why does this matter? Most astronomical systems live in the transition regime ($x \sim 1$–10), not the deep asymptotes. The shape of $\mu(x)$ in this regime determines rotation curve shapes, velocity dispersions, and, crucially, wide binary dynamics.

$\mu(x)$ Form	High-x Behavior	Newtonian Return	Status
Simple (DFP)	$\mu \to 1 - 1/x$	Slow	Derived
Standard	$\mu \to 1 - \frac{1}{2}x^{-2}$	Fast	Fitted

Because $\mu = x/(1+x)$ approaches $\mu \to 1 - 1/x$ as $x \to \infty$, it predicts a slower return to Newtonian behavior than commonly used alternatives. The "standard" form $\mu = x/\sqrt{1+x^2}$ gives $\mu \to 1 - \frac{1}{2}x^{-2}$, which falls off faster. This difference in the quasi-Newtonian tail makes the transition shape testable

in low-acceleration systems such as wide binaries, with the Galactic external field explicitly modeled.

Current observational status: Wide-binary constraints are method-sensitive, with conclusions depending on treatment of hidden triples, selection functions, and velocity modeling. Some analyses favor standard Newtonian/GR behavior; others report anomalies consistent with MOND-like modifications. Wide binaries serve as an active discriminator rather than a settled verdict.

The specific $\mu(x) = x/(1+x)$ derived in Section 3 is not merely "a" function satisfying the asymptotes; it is the unique function within MSC. Its transition shape is therefore a quantitative prediction, not a fitting choice. Future Gaia data releases will test whether wide binaries show the $1/x$ approach (DFP) or the faster $1/x^2$ approach (fitted functions). The data will decide.

8. Interpretation and Limits

Sections 3-4 derived μ(x) and showed it reproduces galactic phenomenology. Section 7 presented the resulting predictions. This section addresses what DFP can and cannot do: its scope as a non-relativistic effective theory, its honest limitations, and the specific tests that could falsify it.

8.1 Summary of Predictions

Before examining scope and limitations, we consolidate what DFP actually predicts. The table below distinguishes three confidence levels: **Theorem** (follows analytically from the postulates), **Model-dependent** (requires additional assumptions about geometry or dynamics), and **Inherited** (carried over from the underlying AQUAL framework).

DFP Predictions: Confidence and Testability

Prediction	Confidence	Uncertainty
BTFR exponent ($v^4 \propto M$)	Theorem	Exact
Phantom density $\rho_{\rm ph} \propto r^{-2}$ (spherical, isolated, deep-MOND exterior)	Theorem	Exact
$\mu(x) = x/(1+x)$ uniqueness	Theorem	Exact
Phantom halo oblateness $q \approx 0.95$	Model-dependent	±0.05
Bar pattern speed $\mathcal{R} \approx 1.0$–1.4	Model-dependent	±0.2
External Field Effect (satellites in hosts)	Inherited from AQUAL	Testable
No intrinsic gravitational response timescale in the present framework	Direct consequence of elliptic/constraint form	Scope statement
Merger lensing: convergence forward-modelable from baryons via AQUAL (no independent DM component)	From operator-residual structure	Testable (quantitative)

Where DFP and ΛCDM Differ

How do DFP predictions compare to the standard paradigm? The table below focuses on **quasi-static, weak-field, galaxy-scale internal dynamics** (rotation curves, scaling relations, low-z phenomenology) where the present DFP/AQUAL framework applies. **Explicitly excluded:** early-universe physics, growth of structure, CMB anisotropies, BAO, and cluster lensing systematics, all of which require relativistic and cosmological completion beyond this paper's scope.

Observational Discriminants: DFP vs. ΛCDM (Galaxy-Scale Dynamics Only)

Observable	DFP Status	ΛCDM Status	Observation / Citation
BTFR $M_b \propto v_{flat}^4$, *tight scatter*	Fundamental scaling in deep-MOND limit (Sec. 2, 3.1)	Achievable in calibrated hydro sims; not a direct analytic prediction of ΛCDM alone	✓ Observed Lelli+ 2016 ApJL 816; McGaugh 2012 AJ 143
Radial Acceleration Relation $g_{obs}(r)$ *tracks* $g_{bar}(r)$	Structural; phantom = Newtonian inversion residual (Sec. 3.2)	Achievable with baryon-halo coupling; not a first-principles law	✓ Observed McGaugh+ 2016 PRL 117; Li+ 2018
Rotation Curve Diversity *at fixed* M_{baryon}	Allowed; geometry/ρ(r) enters directly (AQUAL structure)	Achievable with feedback varying halo concentrations	✓ Observed Oman+ 2015 MNRAS 452
Central Surface Density $\Sigma_\dagger = g_\dagger/(2\pi G)$ *correlation*	Derived from μ(x) structure (Milgrom 2016)	Not a built-in prediction; emerges from baryon-halo interplay	✓ Observed Lelli+ 2016 ApJ 827; Milgrom 2016 PRL 117
Galaxy Cluster Profiles	Requires relativistic completion; underdetermined in present framework	Well-fitted with NFW + hot gas physics	⚠ CDM favored Clowe+ 2006 ApJL 648
Cosmology (CMB, BAO, LSS)	Not addressed; requires relativistic + cosmological completion	Excellent fits with 6-parameter model	⚠ CDM favored Planck 2018
Satellite Plane Anisotropy	No current prediction (N-body extension needed)	Such planes occur in some sims but are uncommon; interpretation debated	? Debated Pawlowski 2018 MPLA 33

Terminology: "Structural/Derived" means the outcome follows from the stated postulates without additional free parameters (section references point to derivations in this paper). "Achievable" means the framework can accommodate the observation with calibrated modeling. "Not addressed" means the present (non-relativistic) DFP framework does not make predictions in that domain. Rows 5-6 are explicit limitations acknowledging where ΛCDM currently has stronger empirical footing.

8.2 Relativistic Extension: The Optical Analog

Sections 3-4 established the constitutive law for galactic dynamics. This section extends DFP to relativistic regimes by deriving the scalar Lagrangian and addressing gravitational lensing through the optical analog framework.

POSTULATE A4: SINGLE-STANDARD CALIBRATION

A field closure boundary provides a **single calibration field** $L(\mathbf{x})$. This calibration defines:

- **Clock calibration:** $d\tau = L\, dt$ (proper time measured by calibrated clocks)
- **Rod calibration:** Spatial distances are defined by radar ranging using the same calibrated clock; proper distance is inferred from round-trip light travel time.

Consequence: This operational definition selects the metric components:

$$A = L^2 \quad \text{(from clock calibration)}$$

$$B = L^{-2} \quad \text{(from radar-ranging rod calibration)}$$

hence $\gamma = 1$ at 1PN.

A4 specifies how rods and clocks are jointly calibrated in a single accounting domain. Both temporal and spatial measurements derive from the same closure output L. Alternative choices $B = f(L)$ with $f \neq L^{-2}$ correspond to a different rod-standard and generically produce gravitational slip ($\gamma \neq 1$). The time coordinate t is adapted to the static Killing field (asymptotic time), so $L = \sqrt{A}$ is the operational redshift factor. This is not runtime routing or message-passing; it is a single consistent calibration field required to avoid competing standards within the same registered domain.

Constraints on Alternatives

A4 is an operational calibration postulate that selects the mapping from the scalar field Φ to observable metric components. It is not derived from a covariant action in this paper; deviations define a "gravitational slip" extension class with testable consequences.

Alternative	Implication	Observational Status
$B \neq L^{-2}$ (different rod standard)	$\gamma \neq 1$ (gravitational slip)	Solar system tests (Cassini) and binary pulsars constrain γ extremely close to 1
Multiple calibration fields	Additional degrees of freedom; generally allows slip unless constrained	More complex; requires extra tuning to satisfy precision tests
No calibration constraint	Metric undetermined	Theory loses predictive content for lensing

A4 is the *minimal* single-field operational calibration that: (1) uses radar-ranging to define spatial distance, (2) employs the same clock for temporal and spatial measurements, and (3) produces $\gamma = 1$ consistent with precision gravitational tests. A4 is a postulate, not a derivation. Departures ($\gamma \neq 1$) constitute a well-defined, testable extension class. A single scalar *can* formally allow $\gamma \neq 1$ if one permits an extra constitutive function $B = f(L)$ with $f \neq L^{-2}$; A4 is the minimal single-function choice. Multi-field relativistic completions (e.g., TeVeS-like theories) can pass EP tests but introduce additional complexity; A4 represents the single-field baseline.

References for γ constraints: Cassini mission: Bertotti et al. 2003, Nature 425, 374. Binary pulsars: Kramer et al. 2021, Phys. Rev. X 11, 041050.

Universal Coupling

A bound composite system (Earth, star, etc.) admits an effective description at scales larger than its internal structure, where only a small set of collective variables (monopole, multipoles, stress-energy moments) source the exterior field. The Interface responds to this effective configuration, not to 10^{50} independent atomic states. All excitations couple to the *same physical metric*. Once the metric sector is fixed by $L(\mathbf{x})$, both massive bodies and light propagate consistently on it. This is the requirement that a single calibration source produces a single geometry. Lensing is therefore forced (at first post-Newtonian order) by single-field calibration closure plus universal coupling.

Shape Locality: A potential concern is whether DFP respects the actual geometry of mass distributions. The answer is built into the formalism. The AQUAL equation

$$\nabla \cdot \left(\mu\left(\frac{|\nabla \Phi|}{a_0} \right) \nabla \Phi \right) = 4\pi G \rho(\mathbf{x})$$

is a *local* partial differential equation. The source $\rho(\mathbf{x})$ is the actual mass distribution (cylinder, sphere, filament, or any other shape). Solving the PDE with this source produces a potential $\Phi(\mathbf{x})$ that naturally respects the geometry of the distribution. The "processing field" is not applied uniformly; it emerges from the actual structure of the source.

Nonlinearity vs. locality: Nonlinearity removes superposition (you cannot add solutions), but it does not remove locality. The field equation $\nabla \cdot (\mu \nabla \Phi) = 4\pi G \rho$ is satisfied *pointwise*. A cylindrical source produces a cylindrical potential not because of "averaging," but because the PDE's differential operator respects the symmetry of the source term locally. This distinction blocks a common MOND/AQUAL misunderstanding.

Metric Form from A4

The clock-rate factor $L(\mathbf{x}) \equiv d\tau/dt$ determines time curvature (standard weak-field GR: $L \simeq 1 + \Phi/c^2$). From Postulate A4 (radar-ranging rod calibration with the same clock), the metric takes the form:

$$ds^2 = -L(\mathbf{x})^2 \, c^2 \, dt^2 + L(\mathbf{x})^{-2} \, d\ell^2$$

1PN Expansion: With $L \simeq 1 + \Phi/c^2$ (weak field):

$$g_{00} = -L^2 \simeq -(1 + 2\Phi/c^2), \quad g_{ij} = L^{-2}\delta_{ij} \simeq (1 - 2\Phi/c^2)\delta_{ij}$$

Hence $\Phi = \Psi$ and $\gamma = 1$ at first post-Newtonian order. This is the standard isotropic weak-field metric with no gravitational slip. This is a **corollary of Postulate A4**, not an independent assumption. The condition $\Phi = \Psi$ ($\gamma = 1$) follows from the operational definition of rod calibration via radar ranging. Departures from A4 would produce $\gamma \neq 1$ and are observationally constrained.

Scope clarification: The 1PN metric structure $g_{00} \approx -(1 + 2\Phi/c^2)$, $g_{ij} \approx (1 - 2\Phi/c^2)\delta_{ij}$ is the standard weak-field form of general relativity. The snapshot calibration axiom identifies Φ with the AQUAL potential rather than the Newtonian potential. Full verification that this identification satisfies the covariant conservation equations $\nabla_\mu T^{\mu\nu} = 0$ and the Bianchi identities requires a complete relativistic theory (e.g., TeVeS or Einstein-Aether extensions), which is beyond the present effective-theory scope.

Constraint on completions: Any relativistic completion of DFP must reduce to the same Φ in the quasi-static weak-field limit and reproduce $\gamma \simeq 1$ (the PPN light-deflection parameter) observationally. This constrains but does not fully determine the UV completion.

Gravitational Lensing (Corollary 1): Lensing is forced by the following causal chain:

1. Solve AQUAL for $\Phi(\mathbf{x})$ given the actual source $\rho(\mathbf{x})$ (shape-respecting because the PDE is local)
2. Map $\Phi \leftrightarrow L$ via the clock bridge: $L \approx 1 + \Phi/c^2$
3. Apply snapshot calibration closure: $A = L^2$, $B = L^{-2}$
4. Light follows null geodesics ($ds^2 = 0$) in this metric, equivalently Fermat's principle with refractive index $n = 1/L^2 \approx 1 - 2\Phi/c^2$

Result: Deflection angle $\alpha = 4GM/(bc^2)$ (GR-strength at 1PN) follows from the closure $B = L^{-2}$, not from any separate "lensing mechanism."

Key point: Everything here is *static + local*. There is no "velocity-reactive medium" or "dynamic response." The field equation is elliptic; the metric closure is algebraic; lensing is geodesic motion in a fixed geometry.

What requires covariant completion: Any prediction involving time evolution of the gravitational field (propagation of changes, response to transients, gravitational waves) lies outside the scope of the present quasi-static framework. As with electrostatics being the static limit of Maxwell's equations, the constraint description here is the quasi-static limit of an underlying causal theory.

Scope lock: Accordingly, this paper makes no claims about gravitational propagation delays, damping, or response timescales.

The Scalar Lagrangian: The Energy Function Behind μ(x)

A relativistic field theory requires an action principle. The "API" of general relativity is the Lagrangian: the energy function that generates the equations of motion. For DFP, we seek the Lagrangian $L(x)$ that produces our constitutive law $\mu(x) = x/(1+x)$.

The relationship between Lagrangian and constitutive law is:

$$\frac{dL}{dx} = x \cdot \mu(x)$$

Substituting $\mu(x) = x/(1+x)$:

$$\frac{dL}{dx} = \frac{x^2}{1+x} = (x-1) + \frac{1}{1+x}$$

Integrating term by term:

$$L(x) = \frac{1}{2}x^2 - x + \ln(1+x)$$

The logarithmic term is striking. It arises naturally from the saturation structure of $\mu(x)$ and suggests an entropic or information-theoretic origin, consistent with the "finite encoding capacity" interpretation of DFP. This Lagrangian is the non-relativistic "engine" that generates our field dynamics.

The Lensing Problem: Asking the Right Question

Scalar field theories have historically predicted half the observed gravitational lensing. In general relativity, light deflection receives equal contributions from time curvature and space curvature, yielding $\theta = 4GM/(bc^2)$. Scalar theories that modify only the potential predict $\theta = 2GM/(bc^2)$. This discrepancy has been treated as a fundamental limitation of MOND-type approaches.

But the standard analysis asks the wrong question. It treats the photon as a passive test particle following geodesics in a pre-existing potential Φ. This framing imports an assumption that DFP does not share: that light is somehow exempt from the physics that governs everything else.

In DFP, a photon is not a particle. It is an electromagnetic field. And in this framework, *every field undergoes closure*. The photon cannot be a passive spectator; it must participate in the same field closure process that governs mass.

The Two-Sphere Interaction: What DFP Requires

This is not an analogy. It is what field closure demands. When a photon traverses a gravitational well, we have two field configurations that must settle at the closure boundary: the gravitational field of the star (coupled to the NET Field) and the electromagnetic field of the photon. Both undergo closure. Both register.

The gravitational field (star + NET Field): The star's mass saturates the vacuum's information capacity, creating a gradient in the background permittivity ε. This is the standard time-dilation component that all scalar theories capture.

The electromagnetic field (photon): The photon is a propagating field configuration. In DFP, it too must couple through closure. For the photon to propagate continuously (no reflection at the saturation boundary), the vacuum's magnetic permeability μ_0 must scale with the permittivity. This is impedance matching: the condition for lossless propagation through a medium with varying electromagnetic properties. (Note: μ_0 here is vacuum permeability, distinct from the constitutive law $\mu(x)$.)

The physical picture: both ε and μ_0 scale by the same saturation factor. The saturation gradient therefore affects the photon's propagation through both its electric and magnetic components. This is why DFP predicts full lensing where passive-particle models predict half.

Mathematical Recovery of the Deflection Angle

Define the saturation factor from the gravitational potential. Following the standard convention in gravitational optics, we write $\Phi = GM/r$ (taking the magnitude; the sign is absorbed into the definition of the refractive index such that $n > 1$ near mass, corresponding to light slowing down):

$$S(r) = 1 + \frac{2\Phi}{c^2} = 1 + \frac{2GM}{rc^2}$$

Under two-sphere saturation, both vacuum constants scale by S. Here ε_0 and μ_0 denote the vacuum permittivity and permeability respectively (standard electromagnetic notation, distinct from the constitutive law $\mu(x)$):

$$\varepsilon = \varepsilon_0 \cdot S, \quad \mu = \mu_0 \cdot S$$

The refractive index is:

$$n = \sqrt{\varepsilon \mu} = \sqrt{\varepsilon_0 \mu_0} \cdot S = S = 1 + \frac{2GM}{rc^2}$$

(normalized to $n = 1$ in vacuum). Applying Fermat's principle of least time, the deflection angle for a light ray with impact parameter b is the path integral of the gradient perpendicular to propagation. For a ray traveling along the z-axis at perpendicular distance $y = b$ from the mass, the gradient is $\partial n / \partial y = -(2GM/c^2) \cdot y / r^3$ where $r = \sqrt{y^2 + z^2}$:

$$\theta = \int_{-\infty}^{+\infty} \left| \frac{\partial n}{\partial y} \right| dz = \frac{2GM}{c^2} \cdot b \int_{-\infty}^{+\infty} \frac{dz}{(b^2 + z^2)^{3/2}}$$

The integral is evaluated by the substitution $z = b \tan \phi$, giving $dz = b \sec^2 \phi \, d\phi$ and $(b^2 + z^2)^{3/2} = b^3 \sec^3 \phi$:

$$\int_{-\infty}^{+\infty} \frac{dz}{(b^2 + z^2)^{3/2}} = \frac{1}{b^2} \int_{-\pi/2}^{\pi/2} \cos \phi \, d\phi = \frac{1}{b^2} \left[\sin \phi \right]_{-\pi/2}^{\pi/2} = \frac{2}{b^2}$$

Therefore:

$$\theta = \frac{2GM}{c^2} \cdot b \cdot \frac{2}{b^2} = \frac{4GM}{bc^2}$$

The final result:

$$\theta = \frac{4GM}{bc^2}$$

This recovers the precise prediction of general relativity. The result is not a coincidence or a parameter fit. It follows from a single requirement: that electromagnetic fields, like all fields in DFP, participate in closure. The "two-sphere" interaction is what field closure demands when light traverses a gravitational well. No tensor metric formulation is required. The full lensing emerges because we asked the right question: not "how does a particle move in a potential?" but "how do two fields close at a common boundary?"

Action Program (Future Work)

What Is Established: The snapshot calibration closure derives the metric form from architectural principles. The AQUAL equation provides the non-relativistic field dynamics. Given $\mu(x)$, one can always construct $F(X)$ with $F'(X) = \mu(\sqrt{X})$ such that the static action reproduces AQUAL.

What Remains: A fully covariant action that:

- Reproduces AQUAL in the static limit ✓ (established)
- Enforces snapshot calibration closure ($A = L^2$, $B = L^{-2}$)
- Couples matter to the physical metric determined by the closure (otherwise universality is lost)
- Avoids pathologies (ghosts, superluminality)

The schematic template is:

$$S = \frac{c^3}{16\pi G}\int \sqrt{-g}\, R \;-\; \frac{a_0^2}{8\pi G}\int \sqrt{-g}\, F(Y) \;+\; S_m[\tilde{g}, \psi]$$

The detailed construction is future work. The present paper establishes the non-relativistic foundation and the calibration closure that any covariant completion must reproduce.

The Cosmological Challenge: CMB and Structure Formation

Scope clarification: DFP is a non-relativistic effective field theory for galactic dynamics. It does not claim to solve cosmology. However, we must address why this is not a fatal omission.

The challenge: The CMB power spectrum (particularly the relative heights of the 2nd and 3rd acoustic peaks) is conventionally interpreted as requiring collisionless dark matter in the early universe. In standard ΛCDM, dark matter provides gravitational potential wells without contributing to the photon-baryon fluid pressure, producing the observed peak ratios. Any modified gravity theory must eventually account for this.

Why standard MOND alone is insufficient: In the early universe, accelerations are high ($g \gg a_0$), so MOND effects are negligible. Yet the CMB "sees" effects that look like dark matter. This is a genuine challenge for MOND-type theories and explains why relativistic completions are essential for cosmology.

Existing progress: Skordis & Złośnik (2021) demonstrated that relativistic MOND completions *can* fit the CMB power spectrum. Their theory (AeST/RMOND) introduces additional fields that become dynamically relevant in the early universe while reducing to AQUAL in the quasi-static limit. The key insight: the fields required to make MOND relativistic can simultaneously source the early-universe effects needed for CMB compatibility. This existence proof shows that CMB success is not exclusive to particle dark matter.

> **What DFP Constrains About Any Completion**
>
> DFP does not solve cosmology, but it constrains what the static limit of any relativistic completion must look like:
>
> - The interpolating function must be $\mu(x) = x/(1+x)$ (derived, not fitted)
> - The architecture must support lossless closure at interfaces
> - The single scale a_0 must emerge from the completion's parameters
>
> Any covariant completion (whether AeST-type, TeVeS-type, or novel) that reproduces the CMB must also reduce to these constraints in the galactic regime. DFP provides the target that cosmological completions must hit.

The research direction: The lensing problem was solved not by adding new particles but by asking the right question: how do fields close at a common boundary? The CMB challenge may require the same shift in perspective. What does field closure predict for gravitational dynamics in a hot, dense, rapidly expanding plasma? The answer lies in extending DFP to cosmological scales, not in postulating new matter species.

Summary: DFP is a non-relativistic effective field theory. Cosmology requires a relativistic extension. Such extensions exist and can fit the CMB (Skordis & Złośnik 2021). DFP constrains what any such completion must satisfy in the galactic regime: the interpolating function $\mu(x) = x/(1+x)$, lossless closure, and the single scale a_0. The path forward is asking better questions about field closure in cosmological contexts, not reverting to particle dark matter.

8.3 Hierarchical Structure

The saturation scale a_0 imposes a universal hierarchy. Every gravitationally coherent source possesses a transition radius $r_c = \sqrt{GM/a_0}$, with the scale depending only on enclosed mass:

Source	Mass	r_c
Sun	$1\,M_\odot$	~7000 AU
Milky Way SMBH	$4 \times 10^6\,M_\odot$	~0.06 pc
Milky Way (total)	$\sim 10^{11}\,M_\odot$	~5 kpc
Rich cluster	$\sim 10^{14}\,M_\odot$	~150 kpc

The scaling $r_c \propto \sqrt{M}$ has a notable consequence: the ratio of interface radius to source size *increases* with mass. A star's r_c is thousands of AU while its physical radius is $\sim 1\,R_\odot$; a galaxy's r_c is comparable to its visible disk. This may explain why MOND effects are subtle in stellar systems but dominant at galactic scales. The interface is far outside the source for stars but comparable to it for galaxies.

8.4 Thermodynamic Viability

Any architecture that processes gravitational effects centrally, maintains a global ledger, or updates microscopic degrees of freedom throughout space faces fundamental thermodynamic constraints. The operation count for cosmic-scale coordination would violate fundamental bounds (cf. Lloyd, 2000). Such an architecture is not merely impractical; it is physically inadmissible. DFP's architecture sidesteps this entirely. Closure is local to each source, requiring no cosmic-scale coordination.

The Locality Principle

A star does not resolve its gravitational relationship to distant objects. It processes only local closure: its mass-energy configuration settling with the NET Field at its boundary, where both exist in the same local conditions. The NET Field at that location already carries the contributions from the rest of the universe, delivered as local boundary data, but crucially, both the star's field object and the NET Field share the same local field state at Σ. The star's "job" is bounded: undergo closure locally, where co-location ensures compatibility, contribute its field outward. It is a pointer, not a processor. Global gravitational structure emerges from this hierarchy of local operations, each thermodynamically bounded, none requiring galactic-scale coordination, because compatibility is ensured by shared local conditions, not global coordination.

The same principle applies at every scale. A moon does not resolve its relationship to distant galaxies or even to other planets in its system. It settles with the NET Field at its own boundary, which already encodes those contributions locally. The moon handles its closure; the universe handles the rest.

And the asteroid from Section 2.5? The same. It encounters the NET Field, "the stream as it is now, here," and both exist in the same local conditions at Σ. They share the same local field state, ensuring compatibility automatically. As the asteroid moves, it continuously settles, not once. The asteroid settles locally: no galactic ledger, no cosmic coordination needed. Just local closure at Σ, where co-location ensures compatibility, repeated everywhere, at every scale. This is why the architecture is thermodynamically viable: no object ever does more than local work, because compatibility comes from shared local conditions, not global coordination.

Distributed by Necessity

This distributed architecture is not imposed on nature; it is how nature operates. Gravitational closure is distributed to each field object, operating on local information at local scales. The cosmic-scale coordination that would be thermodynamically prohibitive simply does not occur. Each source handles its own closure. The universe emerges from the aggregate, not from a master computation.

Settlement as Ontology

The preceding discussion clarifies the architecture: closure is local, distributed, and thermodynamically viable. But what is settlement itself? The answer is ontological, not mechanical.

Two magnetic fields superpose without an external controller computing their sum. The superposition is not a description of what happens; it is what happens. Settlement in DFP has the same status. When fields couple at the boundary Σ, the consistent configuration they reach is not computed by some mechanism running underneath. The settlement is the physical process itself. A useful parallel: quantum systems do not iterate through possibilities linearly; they settle to consistent states through the physics of superposition. DFP's closure operates in the same conceptual register: the answer emerges from the constraint structure, not from sequential processing.

This distinction matters for interpretation. There is no volume-filling update rule, no UV-rate computation, no ledger being maintained. The closure constraint specifies which configurations are admissible; settlement is how fields reach them. Gravity does not require a substrate that calculates gravitational effects. Gravity is what mass does when it closes with the NET Field.

Architectural clarity: The framework requires closure surfaces (standard PDE machinery); the physics is in what closure *does* when capacity saturates (novel). Readers should not interpret Σ as a "new physical thing" added at special locations. It is the enclosing closure surface that any well-posed field theory requires, with the constitutive response $\mu(x)$ encoding the saturation physics that produces MOND phenomenology.

> **Scope: What DFP Predicts for Mergers**
> The quasi-static DFP framework predicts **instantaneous morphology**: given a baryon configuration $\rho_b(\mathbf{x})$, what is the lensing convergence $\kappa(\mathbf{x})$?
>
> DFP does **not** predict in the present framework:
>
> - How fast configurations evolve (baryon dynamics)
> - Gravitational "response times" or "relaxation timescales"
> - Velocity-dependent effects (velocity is not in the field equation)
>
> The tests below focus on the **baryon-only forward model**: can lensing be predicted from observed baryons via AQUAL?

8.5 The Bullet Cluster: Constraint, Not Refutation

> **What Bullet-Cluster Observations Actually Show**
>
> Bullet-like mergers demonstrate that the *lensing convergence* (κ, inferred from shear under standard reconstruction assumptions) can be offset from the X-ray gas centroid. In ΛCDM this is naturally explained by collisionless halos. This is an **existence proof for lensing–gas offsets** in merging clusters, not by itself, a species-level identification of the lensing source.
>
> A common intuition about modified gravity is that "gravity must trace the gas." DFP makes no such assumption. Because the governing relation is nonlinear, the Newtonian inversion residual ("phantom density") can, in principle, peak with compact, higher-acceleration substructure (galaxies) while diffuse gas contributes a broader, lower-amplitude component. Bullet-like systems are therefore **stress-tests of the field-to-lensing mapping** in any effective theory, including DFP. The analytic basis and specific discriminants are developed below.

We treat merging clusters as constraints/discriminants for the framework rather than primary predictions of the present effective theory.

The Bullet Cluster is often cited as definitive evidence against modified gravity. Gravitational lensing reveals mass concentrations offset $\Delta r \approx 150$ kpc from the shocked baryonic gas. This is trivially explained by collisionless dark matter halos passing through each other. We argue this observation is consistent with the DFP framework rather than a refutation of it.

DFP interpretation: A galaxy cluster possesses no single "cluster interface." It is a superposition of the interfaces belonging to its constituent stars and galaxies. In DFP, phantom structure arises from the constitutive relation applied to local field gradients. Compact, interfaced sources (stars, galactic nuclei) produce concentrated gradients and correspondingly sharp phantom peaks. Diffuse gas still sources the AQUAL field and therefore produces lensing, but its smoother distribution yields a broader, lower-contrast phantom component rather than localized peaks.

The key discriminant is **spatial, not temporal**: a baryon-only forward model can be tested against current lensing maps. This is not a future test; it is actionable with existing data. Given observed baryonic distributions (galaxy positions, gas maps), compute the AQUAL potential and derived lensing. Compare to observed convergence maps. Agreement means DFP passes the stress test; disagreement means it fails.

> **Critical Clarification: The Offset Is Spatial, Not Temporal**
>
> The observed offset $\Delta r \approx 150$ kpc arises because phantom density peaks track compact galaxies while diffuse gas produces a broader pedestal. This is not a propagation lag. The DFP field equation is *elliptic* (static, no time derivatives). The potential $\Phi(\mathbf{x})$ is determined instantaneously by the current mass distribution. There is no "response time," no "viscosity," and no "memory" in the field dynamics. The offset is a **spatial structure effect**: compact sources produce steeper field gradients, hence sharper phantom peaks. When clusters collide, galaxies pass through collisionlessly while gas is ram-pressure stripped; the phantom peaks follow the galaxies because that is where the sharp gradients are.

Interpretive Note: Information Load

The following offers an interpretive intuition; the formal account remains the interface-superposition and compact-substructure dominance already described.

The lensing offset can be understood through the lens of information load. The shocked gas at the collision center has high mass density but is spatially diffuse (thermalized plasma with smooth gradients). The stellar/galactic component, though lower in total mass, consists of discrete, spatially concentrated clumps, producing larger spatial gradients $\nabla(|\mathbf{g}|/a_0)$ at fixed total mass.

The field's response (phantom density) reflects this difference in spatial structure. As shown in the Analytic Basis above, the $(\nabla\mu)\cdot\mathbf{g}$ term concentrates residual structure where gradients are sharp. Compact sources produce steeper ∇x, hence sharper phantom peaks that dominate the lensing *peak locations*. The gas still contributes an extended lensing floor (it has mass, therefore it sources the field), but this diffuse component is centered near the X-ray gas centroid. In merging systems, the total lensing map should show: sharp peaks aligned with collisionless substructure (galaxies) plus a broader pedestal near the gas. The *peaks* follow the spatially clumped component, and this is the observed offset, while the gas provides non-zero but lower-contrast convergence.

When two clusters collide, the gas is ram-pressure stripped and concentrated in the collision center, while the galaxies (and their interface hierarchies) pass through each other collisionlessly. The observed lensing offset (*peak* mass concentrations aligned with the galaxies rather than the gas) is consistent with phantom structure tracking the superposed interfaces of compact sources. The gas contributes an extended convergence component centered closer to the X-ray centroid; the peaks track the collisionless substructure. We do not claim this fully explains the microphysics; the observation constrains rather than confirms the framework.

Analytic Basis: Why Residuals Track Compact Substructure

The qualitative "peaks + pedestal" morphology follows from the operator-residual identity derived in Section 4.2:

$$\rho_{\rm ph} = \frac{1}{4\pi G}\nabla \cdot [(\mu - 1)\mathbf{g}]$$

Expanding by the product rule:

$$\rho_{\rm ph} = \frac{1}{4\pi G}\Big[(\mu - 1)\nabla \cdot \mathbf{g} + (\nabla\mu) \cdot \mathbf{g}\Big]$$

The first term tracks the effective-density divergence; the second term, the **sharpening term**, controls spatial variation of the residual. Since $\nabla\mu = (d\mu/dx)\nabla x$ with $x = |\mathbf{g}|/a_0$, and for $\mu(x) = x/(1+x)$ we have $d\mu/dx = 1/(1+x)^2$, this derivative is order-unity for $x \lesssim 1$ and suppressed as $1/x^2$ for $x \gg 1$. Thus spatial variations in μ are suppressed where the local field is everywhere in the Newtonian regime ($x \gg 1$), and remain significant throughout the weak-field regime.

Scaling argument (why compact sources dominate): For a roughly spherical contribution with $g(r) \propto 1/r^2$, we have $x \propto r^{-2}$ and thus

$$|\nabla x| \sim \left|\frac{dx}{dr}\right| \sim \frac{2x}{r}$$

At the same value of x (e.g., in the transition region $x \sim 1$), a *smaller characteristic radius* r yields a proportionally larger $|\nabla x|$, hence larger $|\nabla\mu| \sim (d\mu/dx)|\nabla x|$. Compact substructure therefore produces sharper spatial gradients in μ, concentrating residual structure near galaxies; diffuse gas

produces smoother gradients, contributing a broader floor.

Where spatial variation peaks: For monotone profiles with $g \propto 1/r^2$, the full spatial variation scales as

$$|\nabla\mu| \propto \frac{d\mu}{dx} \cdot |\nabla x| \sim \frac{1}{(1+x)^2} \cdot \frac{2x}{r} = \frac{2}{r} \cdot \frac{x}{(1+x)^2}$$

The function $x/(1+x)^2$ peaks at $x = 1$. This does not contradict the fact that $d\mu/dx$ is monotone decreasing; rather, it shows that *spatial variation* of μ is naturally largest near the transition zone $x \sim 1$ for typical astrophysical profiles. Compact sources reach this transition at smaller radii, yielding higher-contrast projected peaks.

Projection to lensing: Given that the snapshot calibration closure implies $\Phi = \Psi$ at 1PN (Section 8.2), the standard weak-lensing mapping applies: convergence κ tracks the line-of-sight projection of the effective density $\rho_b + \rho_{\rm ph}$. Sharper 3D residual peaks project to sharper 2D convergence peaks. The qualitative "peaks + pedestal" morphology therefore follows from the static operator-residual structure, not from any dynamic or velocity-dependent mechanism.

This is a scaling argument, not a full numerical solve. A quantitative lensing prediction for specific cluster geometries requires solving the nonlinear AQUAL equation for realistic mass distributions, a computation deferred to future work.

Testable Discriminant: Baryon-Only Forward Model

The DFP-native prediction for cluster mergers is that **lensing should be predictable from observed baryons alone**:

> **Core Discriminant**
> Can the observed convergence $\kappa_{\rm obs}(\mathbf{x})$ be forward-modeled from the measured baryon distribution (gas + galaxies) using the AQUAL mapping, **without adding an independent collisionless mass component**?
>
> - **DFP expectation:** Yes. Compute $\kappa_{\rm DFP}$ from $\rho_b(\mathbf{x})$ via AQUAL; residual structure should match the operator-residual / phantom term.
> - **CDM expectation:** No. An independent DM mass map is required to fit $\kappa_{\rm obs}$.

Expected morphology (qualitative): The "peaks + pedestal" structure is expected from the operator-residual identity: compact sources produce sharper spatial gradients in μ, concentrating residual structure near galaxies; diffuse gas produces smoother gradients, contributing a broader floor.

Regarding velocity correlations: Any observed correlation between Δr and $v_{\rm rel}$ is controlled by *baryon dynamics and geometry* (ram-pressure stripping, shock crossing), and is not diagnostic of the gravitational constitutive law by itself.

Proposed Test Protocol

Observables: (1) $\rho_b(\mathbf{x})$: baryon distribution from X-ray (gas) + optical/NIR (galaxies); (2) $\kappa_{\rm obs}(\mathbf{x})$: lensing convergence from weak-lensing reconstruction.

Forward-Model Test: (1) Solve AQUAL for $\Phi(\mathbf{x})$ given $\rho_b(\mathbf{x})$; (2) Compute predicted convergence $\kappa_{\rm DFP}$ from the effective density $\rho_b + \rho_{\rm ph}$; (3) Compare $\kappa_{\rm DFP}$ to $\kappa_{\rm obs}$.

Operational Morphology Metric: Decompose κ into a smoothed ("broad") component and a residual ("sharp") component at scale R. Compare correlations of each component with X-ray gas centroid vs. galaxy light centroid. DFP predicts the sharp component tracks galaxies; the broad component tracks gas.

Candidate Sample: Bullet Cluster (1E 0657-56), El Gordo, Musket Ball, MACS J0025.4-1222, Abell 520, Abell 2744. Note: The ratio $\Delta r/v_{\rm rel}$ is sometimes used as a bookkeeping parameter. In DFP, this ratio has **no interpretation as a gravitational response time**; the effective theory is quasi-static. Any characteristic timescale reflects baryon dynamics.

Quantitative Forward-Model Pipeline (Research Program)

The qualitative "peaks + pedestal" morphology argument requires numerical validation through a rigorous pipeline.

Input Data: X-ray surface brightness provides deprojected gas density $\rho_{\rm gas}(\mathbf{x})$; optical/NIR imaging provides galaxy positions and stellar masses $\rho_*(\mathbf{x})$; combined: $\rho_b(\mathbf{x}) = \rho_{\rm gas}(\mathbf{x}) + \rho_*(\mathbf{x})$.

Boundary Conditions (Critical for AQUAL): AQUAL is nonlinear; the solution depends on boundary data. Operational default: large-domain solve with outer boundary set by an inferred external field $\mathbf{g}_{\rm ext}$ from large-scale structure, validated by sensitivity analysis. Vary both magnitude (±50%) and direction of $\mathbf{g}_{\rm ext}$ within plausible priors to test robustness. Alternative: Dirichlet conditions from interpolated external potential.

Computation: (1) Solve AQUAL equation $\nabla \cdot (\mu \nabla \Phi) = 4\pi G \rho_b$ on 3D adaptive grid; (2) Compute phantom density $\rho_{\rm ph} = (1/4\pi G)\nabla \cdot [(\mu - 1)\mathbf{g}]$; (3) Compute effective 3D density: $\rho_{\rm eff} = \rho_b + \rho_{\rm ph}$.

Lensing Mapping (Under Postulate A4): The mapping $\Phi \to \kappa$ is computed under Postulate A4 ($\Psi = \Phi$, i.e., $\gamma = 1$). Alternative slip choices ($\gamma \neq 1$) define an extension class with different lensing predictions. For cluster lensing, use the thin-lens approximation: (1) Project effective density along

line of sight: $\Sigma_{\rm eff}(x,y) = \int \rho_{\rm eff}(x,y,z)\,dz$; (2) Compute convergence: $\kappa = \Sigma_{\rm eff}/\Sigma_{\rm crit}$, where $\Sigma_{\rm crit} = c^2 D_S/(4\pi G D_L D_{LS})$ is the critical surface density. Use physical (not comoving) surface density and corresponding angular-diameter distances consistently.

Uncertainty Propagation

Baryon Mapping Systematics: Dominant systematics for ρ_b include gas density uncertainties (emissivity modeling, clumping/substructure, triaxiality, absolute calibration), stellar M/L ratio (IMF uncertainty: ±0.2 dex), and boundary condition / $\mathbf{g}_{\rm ext}$ uncertainty.

Lensing Reconstruction Systematics: $\kappa_{\rm obs}$ is not ground truth. Systematics include shear calibration and PSF modeling, source redshift distribution uncertainties, map regularization / mass-sheet degeneracy, and line-of-sight projection / correlated structure. The χ^2 comparison must marginalize over baryon-map nuisance parameters and account for $\kappa_{\rm obs}$ reconstruction uncertainties.

Success Metrics: Peak positions (predicted vs observed centroid offset < 30 kpc); relative peak amplitudes (ratio within 25% of observed); peak-to-pedestal contrast (consistent within uncertainties); aperture mass statistic $M_{\rm ap}$ in matched apertures; multipole moments / quadrupole alignment in matched apertures. These thresholds should be set relative to reconstruction resolution and posteriors; statistical consistency is the criterion.

Target Systems for Baryon-Only Forward Model

Cluster	z	Key Feature	Data Quality
Bullet (1E 0657-56)	0.296	Canonical offset	Excellent
El Gordo	0.87	High-z, high-mass	Good
MACS J0025.4-1222	0.586	Clean bimodal	Good

Scope Boundary: This pipeline tests the quasi-static forward map $\rho_b \mapsto \kappa$ under AQUAL+A4, not the time-dependent merger dynamics. Agreement/failure constrains the static mapping independently of dynamical mechanisms. Existing MOND/AQUAL solvers and multigrid PDE frameworks can be adapted for this pipeline. High-resolution baryon maps from Chandra/XMM (gas) + HST/JWST (stellar) provide input data.

8.6 Ultra-Diffuse Galaxies: An EFE Test Case

The ultra-diffuse galaxies NGC 1052-DF2 and NGC 1052-DF4 have been discussed as potential **EFE-sensitive laboratories**: their low internal accelerations combined with proximity to the massive elliptical NGC 1052 place them in a regime where the External Field Effect should suppress MONDian

behavior, potentially explaining their apparent lack of dark matter.

In MOND/DFP terms: if the external field from NGC 1052 dominates ($g_{\text{ext}} \gtrsim a_0$), then $\mu \approx 1$ throughout the dwarf, and internal dynamics remain near-Newtonian. The "missing" dark matter is not missing; it was never expected in this environment.

Caveat: The status of these systems remains debated due to uncertainties in distance measurements, equilibrium assumptions, and velocity dispersion systematics. We present them as a *discriminant test*, not a confirmed success. If subsequent observations confirm both their proximity to NGC 1052 and their low velocity dispersions, this would constitute strong evidence for the EFE, and against particle dark matter models that predict universal halo-to-stellar-mass ratios regardless of environment.

> **EFE Discriminant**
>
> **DFP/MOND prediction:** Dwarf galaxies in strong external fields ($g_{\text{ext}} \gtrsim a_0$) should exhibit reduced mass discrepancies compared to isolated dwarfs of similar baryonic mass.
>
> **CDM prediction:** No environmental dependence. All dwarfs should have similar dark-to-baryonic mass ratios regardless of host field strength. DF2/DF4 and similar systems provide a natural laboratory for this test.

8.7 Scope and Precision

DFP makes precise claims at a specific level of description. The clarifications below prevent common misreadings and delineate scope.

Physics Clarifications

1. **No superluminal influence.** The field object corresponds to the quantum many-body state of the source; any macroscopic body is a correlated field configuration with intrinsic quantum uncertainty. In the quasi-static regime treated here, the AQUAL field equation is elliptic and should be read as an *instantaneous constraint* selecting self-consistent field configurations given the matter distribution, not as a propagation or signaling law. Accordingly, it introduces no *controllable* superluminal communication channel. Changes in the source propagate causally in any time-dependent completion; this paper does not invoke or require acausal dynamics. (This is analogous to other equal-time constraint structures in field theory, such as the instantaneous Coulomb potential in Coulomb gauge, where causality is preserved in observables.)
2. **No velocity-dependent medium.** There is no wake, no drag, no frictional response in the quasi-static limit. The field equation is elliptic; the closure is algebraic. "Snapshot accounting" means the solution depends on instantaneous configuration, not velocity.
3. **No non-unitary measurement dynamics.** DFP does not invoke collapse, decoherence, or observer-dependent effects. The field closure mechanism is not a measurement process; it is a boundary condition required for well-posed field equations.
4. **Observer-independent closure.** Field closure is a property of the field equations, not a participatory-universe claim.

With these negative statements established, we turn to what DFP positively provides:

What DFP Provides (and What Remains Open)

(5) Effective description delivered; UV mechanism open. DFP provides a complete effective mechanism for field closure: the Field Closure Function role, Full-State Snapshot (snapshot accounting), capacity saturation at a_0, and the derived constitutive response $\mu(x)$. What remains open is the fundamental (UV) completion, which microscopic degrees of freedom (quantum gravity, lossless encoding, etc.) implement this closure. This division is standard: thermodynamics fully describes heat engines without specifying atomic physics. DFP is complete at its level of description.

(6) Geometry matters; charge is conserved. The gravitational field depends on the actual mass distribution $\rho(\mathbf{x})$, not a spherical surrogate: this is standard nonlinear field physics (AQUAL multipole coupling). Coupling load scales with stress-energy, preserving the Equivalence Principle.

(7) Derivation vs. UV origin. DFP derives $\mu(x) = x/(1+x)$ rigorously within the MSC closure class. Whether deeper physics underlies that closure class is a separate UV-completion question, not a weakness of the derivation at the effective level.

(8) The scale a_0 may be cosmological. The proximity $a_0 \sim cH_0$ suggests a cosmological origin via boundary conditions. This connects local dynamics to global structure rather than introducing an ad hoc constant.

(9) Non-relativistic foundation established. DFP provides the quasi-static framework that any relativistic completion must reproduce in the appropriate limit. A covariant extension is therefore constrained in its low-acceleration, quasi-static sector by DFP, even though DFP does not supply a unique relativistic theory.

Terminology Clarification: Rosetta Stone

The following table maps intuitive vocabulary to technical content. The left column provides metaphors for accessibility; the middle column gives formal definitions that the derivation uses; the right column translates to standard physics language. Readers may ignore the metaphors without losing mathematical content.

Intuitive (Metaphor)	DFP Technical Term	EFT/Standard Translation
"The chord"	**Field object**	Correlated many-body field state
"The environment"	**NET Field**	Total gravitational field from external sources
"Where they meet"	**Closure boundary** (Σ)	Geometric locus where constraint is enforced
"The function" / "How they interface"	**Field closure function** (FC_Σ)	Function governing coupling; operates via READ → SETTLE
"Accounting"	**Snapshot accounting** (FSS)	Lossless boundary encoding (what happens inside field closure)
"Coupling load"	Constraint complexity	Information content closure must preserve
"Bookkeeping residual"	Phantom density	Residual from Newtonian inversion of AQUAL solution
"Processing boundary"	r_c contour	Derived threshold where $x = 1$

The technical concepts themselves (field object, field closure, snapshot accounting, phantom density, etc.) are essential to the theory.

8.8 Snapshot Closure: The Field Respects Geometry

A common approximation treats gravitating objects as spheres. For most purposes this is harmless because the far field is dominated by the monopole moment. But DFP makes a stronger claim: **the gravitational field is the closed, conservative solution for the actual configuration, not a spherical surrogate.**

This is not a new postulate; it is what the AQUAL equation already says. The source $\rho(\mathbf{x})$ is the actual mass distribution. The solution $\Phi(\mathbf{x})$ respects that distribution's geometry through the nonlinear constitutive response $\mu(|\nabla\Phi|/a_0)$. Shape and internal structure enter the physics directly.

> **Key Clarification: Closure Is a Condition, Not a Layer**
>
> **Closure is not geometrically "on a surface."** It is logically a closure condition on the *solution*, and therefore applies to the entire coupled configuration. When we say "snapshot accounting respects the actual geometry," we mean: the field is solved for what is actually there, with conservative (no-drag, no-hysteresis) boundary conditions. No new physical layer is introduced. This is a statement about the nature of the closed solution, one that is lossless, history-independent, and configuration-faithful.

This clarification leads to a formal principle that captures DFP's commitment to geometry:

SNAPSHOT CLOSURE PRINCIPLE (DFP)

The gravitational field is the closed, conservative solution for the actual coupled configuration, not a spherical surrogate.

- **"Snapshot accounting"** = no memory, no drag, no hysteresis
- **Closure** is a condition on the solution, not a physical layer
- **Time-lensing** is how the closure cost manifests operationally
- **Shape** enters through $\rho(\mathbf{x})$ and the nonlinear $\mu(|\nabla\Phi|/a_0)$ response

Physical geometry is not approximated. It is what the field is solved for.

This is consistent with standard multipole reasoning: the far field is dominated by the monopole, while geometry enters through higher multipoles and, in DFP, through the nonlinear dependence on $|\nabla\Phi|$. The claim is not that distant observers see detailed structure (they see mostly mass). The claim is that the *closed solution itself* is computed for the actual configuration, not an idealized sphere.

Mathematical Note: Non-Linear Multipole Coupling

In Newtonian gravity, the potential admits a multipole expansion:

$$\Phi_N = \sum_{\ell,m} \frac{M_{\ell m}}{r^{\ell+1}} Y_{\ell m}(\theta,\phi)$$

The linearity of Poisson's equation means monopole, quadrupole, and higher multipoles *decouple*. Each can be computed independently and summed. In AQUAL, the nonlinearity of $\nabla\cdot(\mu\nabla\Phi) = 4\pi G\rho$ **breaks exact decoupling**. Solutions are not additive:

$$\Phi_{\mathrm{AQUAL}}[\rho_1+\rho_2] \neq \Phi_{\mathrm{AQUAL}}[\rho_1] + \Phi_{\mathrm{AQUAL}}[\rho_2]$$

Perturbative expansion: Let $\Phi = \Phi_0 + \delta\Phi$ where Φ_0 is the spherical (monopole) solution. Expanding the nonlinearity:

$$\mu(|\nabla\Phi|) \approx \mu(|\nabla\Phi_0|) + \mu' \frac{\nabla\Phi_0\cdot\nabla\delta\Phi}{|\nabla\Phi_0|}$$

The resulting equation for $\delta\Phi$ has coefficients depending on Φ_0, producing **nonlinear mode coupling**. This is the precise content of "geometry matters": in nonlinear AQUAL, shape information is not captured by simply adding linear multipoles. In the far field, the monopole still dominates, but the nonlinear response introduces **controlled coupling corrections**.

8.9 From Quantum Configuration to Gravitational Field

This section consolidates the physical picture: how the quantum many-body state of matter determines the gravitational field through closure at Σ.

> **Running Example: Why Shape Matters**
>
> Return to our asteroid, but now consider its geometry. It's elongated: 10 km long, 1 km wide, tumbling slowly through space. The same 10^{33} atoms of nickel-iron and silicate, but in an oblong configuration. Compress those atoms into a perfect sphere. Same mass, same composition, same 10^{33} bound fields. **What changes?**
>
> **Far field:** At 1000 km, both shapes produce nearly identical gravitational acceleration. The monopole dominates. **Near field:** At 5 km, the fields diverge. The elongated asteroid has stronger gravity at the ends, weaker along the waist. The sphere has uniform field strength. *Same mass, different field geometry.*
>
> **The key insight:** Stand on the end of the elongated asteroid. Below you: 10 km of rock stretching away, strong gravity. Walk to the waist: only 1 km of rock below, weaker gravity. The field conforms to the actual mass distribution beneath your feet. On a sphere of equal mass? Uniform field strength everywhere. The closure doesn't average this away; it computes the solution for the geometry that actually exists. Field closure registers this difference. When compatibility is enforced at Σ, the closure encodes the actual spatial distribution of stress-energy. The gravitational field is the solution for $\rho(\mathbf{x})$ as it actually is, not a spherical approximation.

> **Snapshot Accounting in Practice**
>
> The gravitational field corresponds to the *current configuration*. No memory of previous shapes. No averaging. The asteroid's geometry enters through $\rho(\mathbf{x})$, and the closed solution tracks it. If the asteroid rotates, tumbles, or fragments, the solution updates immediately for the actual configuration at each moment. This is what "lossless" means operationally: **the field always reflects the current state.**

Three Physical Principles: How Quantum Becomes Real

Here is the deep question: You can *stand* on the asteroid. You can kick it. It will bruise your toe. Yet everything we know about physics says that asteroid is quantum mechanical: 10^{33} atoms, each a quantum field excitation, each in superposition, each entangled with its neighbors. Quantum mechanics doesn't stop at some magic size. There is no "classical transition" where the rules change. The asteroid is as quantum as an electron. So why is it *solid*? Why does it have a definite position? Why does it cast a gravitational shadow you can calculate?

This is the question DFP answers.

Principle 1: The asteroid never stops being quantum. Those 10^{33} atoms are not classical billiard balls. They are a quantum many-body state, a **field object**: correlated, entangled, fundamentally probabilistic. The asteroid *appears* classical because its quantum configuration is dynamically stable, because the correlations lock into patterns that don't slosh around. But the underlying reality remains quantum field theory all the way up.

Principle 2: Field closure is how quantum becomes definite. When the field object couples to the NET Field at the closure boundary Σ, something remarkable happens: the mathematics demands a unique, consistent solution. Not two clock rates at one point. Not contradictory geometries. One settled state B^*_Σ. The closure constraint $\Gamma[\Psi, \Phi_{\mathrm{NET}}, B_\Sigma] = 0$ enforces this. No ambiguity survives. The quantum fuzz *must* resolve into a definite gravitational field.

Principle 3: The gravitational field IS the registration of reality. The settled potential Φ* that emerges from closure is not a secondary effect. It is the mechanism by which quantum configurations become the physical world we measure. Light bends around it. Clocks slow within it. Test masses fall along its gradients. The gravitational field is how 10^{33} quantum excitations become a rock you can hold in your hand.

The Physical Picture: How Solidity Arises

The canonical chain: Field Object + NET Field → Σ → FC_Σ(READ → SETTLE) → B^*_Σ. When fields couple at the closure boundary Σ, the field closure function FC_Σ enforces compatibility: both configurations share the same local geometry $\ell(x)$, enabling a lossless read; the full-state snapshot captures both configurations, and the two fields settle into a unique registered boundary state. Gradients $\nabla\ell(x)$ are gravity.

This is not metaphor. The asteroid's 10^{33} coupled quantum fields do not "approximate" a classical object; they *constitute* one. Solidity, mass, gravitational pull: these are not emergent phenomena layered atop quantum mechanics. They are what field closure *produces* when the consistency condition is satisfied. The quantum configuration remains quantum. What changes is that its gravitational coupling is now registered as a unique, definite field state B*Σ.

This is how classical behavior arises from quantum reality. Not through collapse, not through decoherence, but through the requirement that fields couple consistently. The gravitational field is not a secondary readout of some deeper reality. It *is* the registration of that reality, the unique solution that field closure demands. Gravity is how the universe enforces self-consistency across every boundary, at every moment.

9. Conclusion

For forty years, the interpolating function $\mu(x)$ at the heart of modified gravity was chosen by curve-fitting, not derived from principle. Infinitely many functions satisfy the boundary conditions. In §1.2, we called this the central theoretical gap. This paper closes that gap.

The Key Insight: Why Lossless Closure Determines μ(x)

Why couldn't we derive the correct function? Because we weren't asking what the closure mechanism requires. When matter couples to the gravitational field, multiple contributions must resolve to one consistent state. This is closure. Three constraints are forced by observation:

Lossless: No wakes, drag, or hysteresis are observed; therefore closure preserves information. **Single-channel:** No free parameters to tune; therefore closure operates through one degree of freedom. **Detailed balance:** Gravity is conservative; therefore closure respects equilibrium.

A closure with these properties has exactly one equilibrium form: $\mu(x) = x/(1+x)$. Milgrom's empirical function works because it is the *only* function consistent with how gravitational fields must close. The constitutive law was always there, waiting to be read off the architecture.

With this insight established, the following results emerge:

1. The constitutive law $\mu(x) = x/(1+x)$ is **uniquely derived** as the equilibrium solution of single-channel capacity saturation under detailed balance.
2. The **Baryonic Tully-Fisher Relation** emerges as an asymptotic necessity of the deep-field limit.
3. The phantom "dark matter" density profile $\rho_{\rm ph} \propto r^{-2}$ is an **analytic artifact** of Newtonian inversion.
4. Bar pattern speeds ($\mathcal{R} \approx 1.0$-1.4) distinguish DFP from CDM through reduced dynamical friction.
5. The Bullet Cluster is a **stress-test**: can lensing convergence be predicted from observed baryons via AQUAL without independent dark matter?

Perhaps the deepest implication is geometric: the gravitational field is not computed for an idealized sphere. It is the closed solution for the actual configuration. The sculpture, the galaxy, the person: each has a field solution that respects its geometry through $\rho(\mathbf{x})$ and the nonlinear constitutive response. **Physical geometry is what the closure solves for.**

> **The Complete Framework**
>
> When the **field object** couples to the **NET Field** at the **closure boundary Σ**, **field closure** enforces compatibility and drives **settlement** to a unique **registered boundary state**. This is what AQUAL describes. This is what $\mu(x) = x/(1+x)$ encodes. This is Distributed Field Processing.

The DFP framework suggests that "dark matter" phenomena arise not from hidden particles, but from the informational structure of the gravitational field itself. What observers call "dark matter" is simply the field's memory of its sources.

From galactic dynamics to relativistic extension. This paper establishes DFP as a non-relativistic effective theory, but Section 8 develops the bridge toward full general relativity. The calibration closure (Postulate A4) specifies how the scalar potential maps to spacetime geometry: clocks slow, rulers shrink, and light bends according to a single consistent calibration field. This optical analog framework provides constraints for relativistic completions. Whether the UV theory is TeVeS-type, AeST-type, or something novel, consistency with galactic observations requires recovering $\mu(x) = x/(1+x)$ in the quasi-static limit, supporting lossless closure, and producing the single scale a_0.

What DFP offers. For galactic dynamics (rotation curves, bar pattern speeds, lensing), the framework is directly applicable. For cosmology (CMB, structure formation), relativistic completions like Skordis and Złośnik's AeST have demonstrated that MOND-type theories can fit the power spectrum; DFP identifies what such completions should recover in the galactic regime. The predictions in Section 7 and the Bullet Cluster pipeline in Section 8 provide concrete tests. Observation will adjudicate.

Distributed Field Processing contributes to the broader study of **field dynamics**, the systematic investigation of how fields mediate interactions under informational and physical constraints. The principles explored here (finite encoding capacity, lossless closure, local mediation) may extend beyond gravitation. DFP is one step in that direction.

Acknowledgments

We stand on the shoulders of giants. Einstein taught us that "there is no space empty of field" and that geometry itself responds to matter. Feynman showed how to think clearly about quantum fields and demanded that physics be explicable to anyone willing to engage. Hawking connected gravity to information in ways we are still learning to appreciate. Their work shapes every page of this paper.

This work builds directly upon the contributions of M. Milgrom, who introduced MOND and recognized that galactic dynamics encode something fundamental, and J.D. Bekenstein, who with Milgrom developed the AQUAL formulation that provides the mathematical framework here. The author is indebted to the observational work of S.S. McGaugh, B. Famaey, F. Lelli, and collaborators, whose precise measurements of galactic dynamics and the radial acceleration relation motivated this theoretical investigation.

The closure architecture developed here draws on standard quantum field theory: matter as correlated field configurations, consistency requirements at boundaries, and the constraint structure of effective theories. The author thanks the broader MOND research community for decades of work that made this synthesis possible.

Finally, thanks to the sun for shining.

Appendix A: Formal Lemmas (Proof Spine)

This appendix formalizes the proof sketches in §3 into short lemmas. Throughout, let $x \equiv |\nabla\Phi|/a_0$ denote the dimensionless acceleration ratio, and let $\mu(x)$ be the constitutive factor in the AQUAL equation.

> **Assumptions (Minimal Single-Channel Closure)**
>
> The following lemmas assume the MSC class defined in §3.1:
>
> - **MSC1:** Single occupancy variable $\varepsilon \in [0, 1]$
> - **MSC2:** Forward rate $\propto x^n$ and available capacity $(1 - \varepsilon)$
> - **MSC3:** Reverse rate $\propto \varepsilon$ (single-channel relaxation)
> - **MSC4:** Constitutive identification $\mu(x) \equiv \varepsilon_{\text{eq}}(x)$
>
> **Scope:** Theorem A.5 proves uniqueness *within MSC*, not uniqueness among all possible closure mechanisms.

Lemma A.1 (Detailed-Balance Equilibrium)

Assume: A single occupancy variable $\varepsilon \in [0,1]$ with reversible transitions between "unresolved" $(1-\varepsilon)$ and "resolved" (ε) states, with rates

$$r_+(x,\varepsilon) = \alpha x^n (1-\varepsilon), \quad r_-(\varepsilon) = \beta\varepsilon$$

for some $\alpha, \beta > 0$ and $n > 0$.

Then: The unique equilibrium satisfies

$$\varepsilon_{\mathrm{eq}}(x) = \frac{x^n}{k + x^n}, \quad k \equiv \beta/\alpha.$$

> *Proof.* Detailed balance requires $r_+ = r_-$, giving $\alpha x^n(1-\varepsilon) = \beta\varepsilon$. Solving for ε yields $\varepsilon = \alpha x^n/(\alpha x^n + \beta) = x^n/(x^n + k)$. Uniqueness follows because the balance equation is linear in ε with $\alpha, \beta > 0$. ■

Lemma A.2 (Deep-MOND Scaling Forces $n = 1$)

Assume: The constitutive identification $\mu(x) \equiv \varepsilon_{\mathrm{eq}}(x)$, and the empirical deep-MOND boundary condition $\mu(x) \sim x$ as $x \to 0$ (Postulate 2).

Then: $n = 1$, and k is fixed by the choice of a_0.

> *Proof.* From Lemma A.1, as $x \to 0$: $\mu(x) = x^n/(k + x^n) \approx x^n/k$. The deep-MOND condition $\mu(x) \sim x$ forces $n = 1$. The remaining factor $1/k$ fixes the normalization of x, equivalently a rescaling of a_0, so one may set $k = 1$ by definition of the acceleration scale. ■

Lemma A.3 (Unique μ in MSC)

Under the hypotheses of Lemmas A.1–A.2, $\mu(x)$ is uniquely determined:

$$\mu(x) = \frac{x}{1+x}.$$

> *Proof.* Lemma A.1 gives $\mu(x) = x^n/(k + x^n)$. Lemma A.2 forces $n = 1$ and $k = 1$ (by normalization). Substitute to obtain $\mu(x) = x/(1 + x)$. ■

Lemma A.4 (Newtonian Correspondence)

The $\mu(x)$ in Lemma A.3 satisfies the Newtonian limit (Postulate 1): $\mu(x) \to 1$ as $x \to \infty$.

> *Proof.* From $\mu(x) = x/(1 + x)$: as $x \to \infty$, $\mu \to x/x = 1$. ■

Theorem A.5 (Constitutive Law in MSC)

In the Minimal Single-Channel Saturating Closure (MSC) class defined in §3.1, the interpolating function is uniquely determined as

$$\mu(x) = \frac{x}{1+x}.$$

> *Proof.* Lemma A.1 establishes the equilibrium form under detailed balance. Lemma A.2 fixes $n = 1$ via the deep-MOND boundary condition. Lemma A.3 yields the explicit μ. Lemma A.4 verifies the Newtonian limit. The result follows. ■

Dependency graph: A.1 → A.2 → A.3 → A.4 → A.5 (Theorem). Each lemma proves exactly one step; the main theorem chains them.

References

[1] Bekenstein, J. D. & Milgrom, M. "Does the missing mass problem signal the breakdown of Newtonian gravity?" *Astrophys. J.* **286**, 7–14 (1984). doi:10.1086/162570

[2] Milgrom, M. "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis." *Astrophys. J.* **270**, 365–370 (1983). doi:10.1086/161130

[3] Famaey, B. & McGaugh, S. S. "Modified Newtonian Dynamics (MOND): Observational Phenomenology and Relativistic Extensions." *Living Rev. Relativ.* **15**, 10 (2012). doi:10.12942/lrr-2012-10

[4] Famaey, B. & Binney, J. "Modified Newtonian dynamics in the Milky Way." *Mon. Not. R. Astron. Soc.* **363**, 603–608 (2005). doi:10.1111/j.1365-2966.2005.09474.x

[5] Bekenstein, J. D. "Universal upper bound on the entropy-to-energy ratio for bounded systems." *Phys. Rev. D* **23**, 287–298 (1981). doi:10.1103/PhysRevD.23.287

[6] Gentile, G., Famaey, B. & de Blok, W. J. G. "THINGS about MOND." *Astron. Astrophys.* **527**, A76 (2011). doi:10.1051/0004-6361/201015283

[7] Clowe, D. et al. "A Direct Empirical Proof of the Existence of Dark Matter." *Astrophys. J. Lett.* **648**, L109–L113 (2006). doi:10.1086/508162

[8] Chae, K.-H. et al. "Robust Evidence for the Breakdown of Standard Gravity at Low Acceleration from Statistically Pure Binaries." *Astrophys. J.* **952**, 128 (2023). doi:10.3847/1538-4357/ace101

[9] Roshan, M. et al. "Fast galaxy bars continue to challenge standard cosmology." *Mon. Not. R. Astron. Soc.* **508**, 926–939 (2021). doi:10.1093/mnras/stab2553

[10] Banik, I. & Zhao, H. "From Galactic Bars to the Hubble Tension: Weighing Up the Astrophysical Evidence for Milgromian Gravity." *Symmetry* **14**, 1331 (2022). doi:10.3390/sym14071331

[11] Abbott, B. P. et al. (LIGO Scientific Collaboration and Virgo Collaboration). "Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A." *Astrophys. J. Lett.* **848**, L13 (2017). doi:10.3847/2041-8213/aa91c9

[12] Bousso, R. "The holographic principle." *Rev. Mod. Phys.* **74**, 825 (2002). doi:10.1103/RevModPhys.74.825

[13] Skordis, C. & Złośnik, T. "New Relativistic Theory for Modified Newtonian Dynamics." *Phys. Rev. Lett.* **127**, 161302 (2021). doi:10.1103/PhysRevLett.127.161302

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